Neural Computation -- 2011

120 article(s), all issues

1. Revisiting tests for neglected nonlinearity using artificial neural networks

ChoJin Seo, IshidaIsao, WhiteHalbert

https://doi.org/10.5555/2646468.2646470

Tests for regression neglected nonlinearity based on artificial neural networks ANNs have so far been studied by separately analyzing the two ways in which the null of regression linearity can hold...

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2. Multistability and Perceptual Inference

Samuel J. Gershman, Edward Vul, Joshua B. Tenenbaum

https://doi.org/10.1162/neco_a_00226 | pp. 1-24

Ambiguous images present a challenge to the visual system: How can uncertainty about the causes of visual inputs be represented when there are multiple equally plausible causes? A Bayesian ideal observer should represent uncertainty in the form of a posterior probability distribution over causes. However, in many real-world situations, computing this distribution is intractable and requires some form of approximation. We argue that the visual system approximates the posterior over underlying causes with a set of samples and that this approximation strategy produces perceptual multistability--stochastic alternation between percepts in consciousness. Under our analysis, multistability arises from a dynamic sample-generating process that explores the posterior through stochastic diffusion, implementing a rational form of approximate Bayesian inference known as Markov chain Monte Carlo (MCMC). We examine in detail the most extensively studied form of multistability, binocular rivalry, showing how a variety of experimental phenomena--gamma-like stochastic switching, patchy percepts, fusion, and traveling waves--can be understood in terms of MCMC sampling over simple graphical models of the underlying perceptual tasks. We conjecture that the stochastic nature of spiking neurons may lend itself to implementing sample-based posterior approximations in the brain.

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3. Mathematical Equivalence of Two Common Forms of Firing Rate Models of Neural Networks

Kenneth D. Miller, Francesco Fumarola

https://doi.org/10.1162/neco_a_00221 | pp. 25-31

We demonstrate the mathematical equivalence of two commonly used forms of firing rate model equations for neural networks. In addition, we show that what is commonly interpreted as the firing rate in one form of model may be better interpreted as a low-pass-filtered firing rate, and we point out a conductance-based firing rate model.

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4. Interactions of Excitatory and Inhibitory Feedback Topologies in Facilitating Pattern Separation and Retrieval

Jane X. Wang, Micha■ ■ochowski

https://doi.org/10.1162/neco_a_00220 | pp. 32-59

Within the brain, the interplay between connectivity patterns of neurons and their spatiotemporal dynamics is believed to be intricately linked to the bases of behavior, such as the process of storing, consolidating, and retrieving memory traces. Memory is believed to be stored in the synaptic patterns of anatomical circuitry in the form of increased connectivity densities within subpopulations of neurons. At the same time, memory recall is thought to correspond to activation of discrete areas of the brain corresponding to those memories. Such regional subpopulations can selectively activate during memory recall or retrieval, signifying the process of accessing a single memory or concept. It has been shown previously that recovery of single memory activity patterns is mediated by global neuromodulation signifying transition into different cognitive states such as sleep or awake exploration. We examine how underlying topology can affect memory awake activation and sleep reactivation when such memories share increasing proportions of neurons. The results show that while single memory activation is diminished with increased overlap, pattern separation can be recovered by offsetting excitatory associations between two memories with targeted and heterogeneous inhibitory feedback. Such findings point to the importance of excitatory-to-inhibitory current balance at both the global and local levels in the context of memory retrieval and replay, and highlight the role of network topology in memory management processes.

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5. Unsupervised Learning of Generative and Discriminative Weights Encoding Elementary Image Components in a Predictive Coding Model of Cortical Function

Michael Spratling

https://doi.org/10.1162/neco_a_00222 | pp. 60-103

A method is presented for learning the reciprocal feedforward and feedback connections required by the predictive coding model of cortical function. When this method is used, feedforward and feedback connections are learned simultaneously and independently in a biologically plausible manner. The performance of the proposed algorithm is evaluated by applying it to learning the elementary components of artificial and natural images. For artificial images, the bars problem is employed, and the proposed algorithm is shown to produce state-of-the-art performance on this task. For natural images, components resembling Gabor functions are learned in the first processing stage, and neurons responsive to corners are learned in the second processing stage. The properties of these learned representations are in good agreement with neurophysiological data from V1 and V2. The proposed algorithm demonstrates for the first time that a single computational theory can explain the formation of cortical RFs and also the response properties of cortical neurons once those RFs have been learned.

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6. Recurrent Kernel Machines: Computing with Infinite Echo State Networks

Michiel Hermans, Benjamin Schrauwen

https://doi.org/10.1162/neco_a_00200 | pp. 104-133

Echo state networks (ESNs) are large, random recurrent neural networks with a single trained linear readout layer. Despite the untrained nature of the recurrent weights, they are capable of performing universal

computations on temporal input data, which makes them interesting for both theoretical research and practical applications. The key to their success lies in the fact that the network computes a broad set of nonlinear, spatiotemporal mappings of the input data, on which linear regression or classification can easily be performed. One could consider the reservoir as a spatiotemporal kernel, in which the mapping to a high-dimensional space is computed explicitly. In this letter, we build on this idea and extend the concept of ESNs to infinite-sized recurrent neural networks, which can be considered recursive kernels that subsequently can be used to create recursive support vector machines. We present the theoretical framework, provide several practical examples of recursive kernels, and apply them to typical temporal tasks.

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7. A Scale-Invariant Internal Representation of Time

Karthik H. Shankar, Marc W. Howard

https://doi.org/10.1162/neco_a_00212 | pp. 134-193

We propose a principled way to construct an internal representation of the temporal stimulus history leading up to the present moment. A set of leaky integrators performs a Laplace transform on the stimulus function, and a linear operator approximates the inversion of the Laplace transform. The result is a representation of stimulus history that retains information about the temporal sequence of stimuli. This procedure naturally represents more recent stimuli more accurately than less recent stimuli; the decrement in accuracy is precisely scale invariant. This procedure also yields time cells that fire at specific latencies following the stimulus with a scale-invariant temporal spread. Combined with a simple associative memory, this representation gives rise to a moment-to-moment prediction that is also scale invariant in time. We propose that this scale-invariant representation of temporal stimulus history could serve as an underlying representation accessible to higher-level behavioral and cognitive mechanisms. In order to illustrate the potential utility of this scale-invariant representation in a variety of fields, we sketch applications using minimal performance functions to problems in classical conditioning, interval timing, scale-invariant learning in autoshaping, and the persistence of the recency effect in episodic memory across timescales.

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8. Geometry-Invariant Texture Retrieval Using a Dual-Output Pulse-Coupled Neural Network

Xiaojun Li, Yide Ma, Zhaobin Wang, Wenrui Yu

https://doi.org/10.1162/neco_a_00194 | pp. 194-216

This letter proposes a novel dual-output pulse coupled neural network model (DPCNN). The new model is applied to obtain a more stable texture description in the face of the geometric transformation. Time series, which are computed from output binary images of DPCNN, are employed as translation-, rotation-, scale-, and distortion-invariant texture features. In the experiments, DPCNN has been well tested by using Brodatz's album and the VisTex database. Several existing models are compared with the proposed DPCNN model. The experimental results, based on different testing data sets for images with different translations, orientations, scales, and affine transformations, show that our proposed model outperforms existing models in geometry-invariant texture retrieval. Furthermore, the robustness of DPCNN to noisy data is examined in the experiments.

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9. Guaranteed Cost Synchronization of Chaotic Cellular Neural Networks with Time-Varying Delay

Jianjun Tu, Hanlin He

https://doi.org/10.1162/neco_a_00191 | pp. 217-233

Synchronization of cellular neural networks with time-varying delay is discussed in this letter. Based on Razumikhin theorem, a guaranteed cost synchronous controller is given. Unlike Lyapunov-Krasovskii analysis process, there is no constraint on the change rate of time delay. The saturated terms emerging in the Razumikhin analysis are amplified by zoned discussion and maximax synthesis rather than by Lipschitz condition and vector inequality, which will bring more conservatism. Then the controller criterion is transformed from quadratic matrix inequality form into linear matrix inequality form, with the help of a sufficient and necessary transformation condition. The minimization of the guaranteed cost is studied, and a further criterion for getting the controller is presented. Finally, the guaranteed cost synchronous control and its corresponding minimization problem are illustrated with examples of chaotic time-varying delay cellular neural networks.

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10. Efficient Blind Dereverberation and Echo Cancellation Based on Independent Component Analysis for Actual Acoustic Signals

Ryu Takeda, Kazuhiro Nakadai, Tôru Takahashi, Kazunori Komatani, Tetsuya Ogata, Hiroshi G. Okuno

https://doi.org/10.1162/neco_a_00219 | pp. 234-272

This letter presents a new algorithm for blind dereverberation and echo cancellation based on independent component analysis (ICA) for actual acoustic signals. We focus on frequency domain ICA (FD-ICA) because its computational cost and speed of learning convergence are sufficiently reasonable for practical applications such as hands-free speech recognition. In applying conventional FD-ICA as a preprocessing of automatic speech recognition in noisy environments, one of the most critical problems is how to cope with reverberations. To extract a clean signal from the reverberant observation, we model the separation process in the short-time Fourier transform domain and apply the multiple input/output inverse-filtering theorem (MINT) to the FD-ICA separation model. A naive implementation of this method is computationally expensive, because its time complexity is the second order of reverberation time. Therefore, the main issue in dereverberation is to reduce the high computational cost of ICA. In this letter, we reduce the computational complexity to the linear order of the reverberation time by using two techniques: (1) a separation model based on the independence of delayed observed signals with MINT and (2) spatial sphering for preprocessing. Experiments show that the computational cost grows in proportion to the linear order of the reverberation time and that our method improves the word correctness of automatic speech recognition by 10 to 20 points in a RT■■= 670 ms reverberant environment.

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11. Higher-Order Approximations for Testing Neglected Nonlinearity

Halbert White, Jin Seo Cho

https://doi.org/10.1162/neco_a_00225 | pp. 273-287

We illustrate the need to use higher-order (specifically sixth-order) expansions in order to properly determine the asymptotic distribution of a standard artificial neural network test for neglected nonlinearity. The test

statistic is a quasi-likelihood ratio (QLR) statistic designed to test whether the mean square prediction error improves by including an additional hidden unit with an activation function violating the no-zero condition in Cho, Ishida, and White (2011). This statistic is also shown to be asymptotically equivalent under the null to the Lagrange multiplier (LM) statistic of Luukkonen, Saikkonen, and Teräsvirta (1988) and Teräsvirta (1994). In addition, we compare the power properties of our QLR test to one satisfying the no-zero condition and find that the latter is not consistent for detecting a DGP with neglected nonlinearity violating an analogous no-zero condition, whereas our QLR test is consistent.

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12. How the Brain Generates Movement

Uri Rokni, Haim Sompolinsky

https://doi.org/10.1162/neco_a_00223 | pp. 289-331

In this study, we assume that the brain uses a general-purpose pattern generator to transform static commands into basic movement segments. We hypothesize that this pattern generator includes an oscillator whose complete cycle generates a single movement segment. In order to demonstrate this hypothesis, we construct an oscillator-based model of movement generation. The model includes an oscillator that generates harmonic outputs whose frequency and amplitudes can be modulated by external inputs. The harmonic outputs drive a number of integrators, each activating a single muscle. The model generates muscle activation patterns composed of rectilinear and harmonic terms. We show that rectilinear and fundamental harmonic terms account for known properties of natural movements, such as the invariant bell-shaped hand velocity profile during reaching. We implement these dynamics by a neural network model and characterize the tuning properties of the neural integrator cells, the neural oscillator cells, and the inputs to the system. Finally, we propose a method to test our hypothesis that a neural oscillator is a central component in the generation of voluntary movement.

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13. Noise Tolerance of Attractor and Feedforward Memory Models

Sukbin Lim, Mark S. Goldman

https://doi.org/10.1162/neco_a_00234 | pp. 332-390

In short-term memory networks, transient stimuli are represented by patterns of neural activity that persist long after stimulus offset. Here, we compare the performance of two prominent classes of memory networks, feedback-based attractor networks and feedforward networks, in conveying information about the amplitude of a briefly presented stimulus in the presence of gaussian noise. Using Fisher information as a metric of memory performance, we find that the optimal form of network architecture depends strongly on assumptions about the forms of nonlinearities in the network. For purely linear networks, we find that feedforward networks outperform attractor networks because noise is continually removed from feedforward networks when signals exit the network; as a result, feedforward networks can amplify signals they receive faster than noise accumulates over time. By contrast, attractor networks must operate in a signal-attenuating regime to avoid the buildup of noise. However, if the amplification of signals is limited by a finite dynamic range of neuronal responses or if noise is reset at the time of signal arrival, as suggested by recent experiments, we find that attractor networks can outperform feedforward ones. Under a simple model in which neurons have a finite dynamic range, we find that the optimal attractor networks are forgetful if there is no mechanism for noise reduction with signal arrival but nonforgetful (perfect integrators) in the presence of

a strong reset mechanism. Furthermore, we find that the maximal Fisher information for the feedforward and attractor networks exhibits power law decay as a function of time and scales linearly with the number of neurons. These results highlight prominent factors that lead to trade-offs in the memory performance of networks with different architectures and constraints, and suggest conditions under which attractor or feedforward networks may be best suited to storing information about previous stimuli.

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14. Transmission of Population-Coded Information

Alfonso Renart, Mark C. W. van Rossum

https://doi.org/10.1162/neco_a_00227 | pp. 391-407

As neural activity is transmitted through the nervous system, neuronal noise degrades the encoded information and limits performance. It is therefore important to know how information loss can be prevented. We study this question in the context of neural population codes. Using Fisher information, we show how information loss in a layered network depends on the connectivity between the layers. We introduce an algorithm, reminiscent of the water filling algorithm for Shannon information that minimizes the loss. The optimal connection profile has a center-surround structure with a spatial extent closely matching the neurons' tuning curves. In addition, we show how the optimal connectivity depends on the correlation structure of the trial-to-trial variability in the neuronal responses. Our results explain how optimal communication of population codes requires the center-surround architectures found in the nervous system and provide explicit predictions on the connectivity parameters.

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15. Quantifying Statistical Interdependence, Part III: N > 2 Point Processes

Justin Dauwels, Théophane Weber, François Vialatte, Toshimitsu Musha, Andrzej Cichocki

https://doi.org/10.1162/neco_a_00235 | pp. 408-454

Stochastic event synchrony (SES) is a recently proposed family of similarity measures. First, "events" are extracted from the given signals; next, one tries to align events across the different time series. The better the alignment, the more similar the N time series are considered to be. The similarity measures quantify the reliability of the events (the fraction of "nonaligned" events) and the timing precision. So far, SES has been developed for pairs of one-dimensional (Part I) and multidimensional (Part II) point processes. In this letter (Part III), SES is extended from pairs of signals to N > 2 signals. The alignment and SES parameters are again determined through statistical inference, more specifically, by alternating two steps: (1) estimating the SES parameters from a given alignment and (2), with the resulting estimates, refining the alignment. The SES parameters are computed by maximum a posteriori (MAP) estimation (step 1), in analogy to the pairwise case. The alignment (step 2) is solved by linear integer programming. In order to test the robustness and reliability of the proposed N-variate SES method, it is first applied to synthetic data. We show that N-variate SES results in more reliable estimates than bivariate SES. Next N-variate SES is applied to two problems in neuroscience: to quantify the firing reliability of Morris-Lecar neurons and to detect anomalies in EEG synchrony of patients with mild cognitive impairment. Those problems were also considered in Parts I and II, respectively. In both cases, the N-variate SES approach yields a more detailed analysis.

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16. Cross-Talk Induces Bifurcations in Nonlinear Models of Synaptic Plasticity

Terry Elliott

https://doi.org/10.1162/neco_a_00224 | pp. 455-522

Linear models of synaptic plasticity provide a useful starting-point for examining the dynamics of neuronal development and learning, but their inherent problems are well known. Models of synaptic plasticity that embrace the demands of biological realism are therefore typically nonlinear. Viewed from a more abstract perspective, nonlinear models of synaptic plasticity are a subset of nonlinear dynamical systems. As such, they may therefore exhibit bifurcations under the variation of control parameters, including noise and errors in synaptic updates. One source of noise or error is the cross-talk that occurs during otherwise Hebbian plasticity. Under cross-talk, stimulation of a set of synapses can induce or modify plasticity in adjacent, unstimulated synapses. Here, we analyze two nonlinear models of developmental synaptic plasticity and a model of independent component analysis in the presence of a simple model of cross-talk. We show that cross-talk does indeed induce bifurcations in these models, entirely destroying their ability to acquire either developmentally or learning-related patterns of fixed points. Importantly, the critical level of cross-talk required to induce bifurcations in these models is very sensitive to the statistics of the afferents' activities and the number of afferents synapsing on a postsynaptic cell. In particular, the critical level can be made arbitrarily small. Because bifurcations are inevitable in nonlinear models, our results likely apply to many nonlinear models of synaptic plasticity, although the precise details vary by model. Hence, many nonlinear models of synaptic plasticity are potentially fatally compromised by the toxic influence of cross-talk and other sources of noise and errors more generally. We conclude by arguing that biologically realistic models of synaptic plasticity must be robust against noise-induced bifurcations and that biological systems may have evolved strategies to circumvent their possible dangers.

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17. Intrinsic Adaptation in Autonomous Recurrent Neural Networks

Dimitrije Markovi■, Claudius Gros

https://doi.org/10.1162/neco_a_00232 | pp. 523-540

A massively recurrent neural network responds on one side to input stimuli and is autonomously active, on the other side, in the absence of sensory inputs. Stimuli and information processing depend crucially on the quality of the autonomous-state dynamics of the ongoing neural activity. This default neural activity may be dynamically structured in time and space, showing regular, synchronized, bursting, or chaotic activity patterns. We study the influence of nonsynaptic plasticity on the default dynamical state of recurrent neural networks. The nonsynaptic adaption considered acts on intrinsic neural parameters, such as the threshold and the gain, and is driven by the optimization of the information entropy. We observe, in the presence of the intrinsic adaptation processes, three distinct and globally attracting dynamical regimes: a regular synchronized, an overall chaotic, and an intermittent bursting regime. The intermittent bursting regime is characterized by intervals of regular flows, which are quite insensitive to external stimuli, interceded by chaotic bursts that respond sensitively to input signals. We discuss these findings in the context of self-organized information processing and critical brain dynamics.

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18. Kernel Current Source Density Method

Jan Potworowski, Wit Jakuczun, Szymon ■■ski, Daniel K. Wójcik
https://doi.org/10.1162/neco_a_00236 | pp. 541-575

Local field potentials (LFP), the low-frequency part of extracellular electrical recordings, are a measure of the neural activity reflecting dendritic processing of synaptic inputs to neuronal populations. To localize synaptic dynamics, it is convenient, whenever possible, to estimate the density of transmembrane current sources (CSD) generating the LFP. In this work, we propose a new framework, the kernel current source density method (kCSD), for nonparametric estimation of CSD from LFP recorded from arbitrarily distributed electrodes using kernel methods. We test specific implementations of this framework on model data measured with one-, two-, and three-dimensional multielectrode setups. We compare these methods with the traditional approach through numerical approximation of the Laplacian and with the recently developed inverse current source density methods (iCSD). We show that iCSD is a special case of kCSD. The proposed method opens up new experimental possibilities for CSD analysis from existing or new recordings on arbitrarily distributed electrodes (not necessarily on a grid), which can be obtained in extracellular recordings of single unit activity with multiple electrodes.

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19. Erratum (“Feature selection in simple neurons: how coding depends on spiking dynamics” by Michael Famulare and Adrienne Fairhall, Neural Computation, February 2012, Vol. 22, No. 3: 581–598)

(authors unknown)
https://doi.org/10.1162/neco_c_00276 | pp. 576-576

The text from the start of Section 4.6 through equation 4.10 should read: To find the STA for the linear model and compare it to numerical simulations, we move to discrete time by defining t=nh, where n is an integer and h is the time step. For clarity of notation, we identify v(t)=v(nh) ≡ vn. The linear model in equation 4.3 is equivalently described by the forward transition probability distribution: 4.10cAlso of use are the steady-state probability distribution, pL(v), given in equation 4.4, and the backward transition probability distribution, p(vn|vn+1), which can be derived with Bayes’ rule: 4.9c for small . Notice that the linear model is statistically reversible (Weiss, 1975): the backward transition distribution is the time reversal of the forward, vn ■ vn+1.With corrected equation 4.9c, equation 4.19 becomes: 4.19cFor small , the correction amounts to a sign change, h→ -h. Despite the correction to equation 4.19, equation 4.20 in the original paper and the results that follow from it are correct because there was an additional minus sign error hidden in the transition from equation 4.19 to equation 4.20. Like good students, we made an even number of errors.

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20. MOSAIC for Multiple-Reward Environments

Norikazu Sugimoto, Masahiko Haruno, Kenji Doya, Mitsuo Kawato
https://doi.org/10.1162/neco_a_00246 | pp. 577-606

Reinforcement learning (RL) can provide a basic framework for autonomous robots to learn to control and maximize future cumulative rewards in complex environments. To achieve high performance, RL controllers must consider the complex external dynamics for movements and task (reward function) and optimize control commands. For example, a robot playing tennis and squash needs to cope with the different dynamics of a tennis or squash racket and such dynamic environmental factors as the wind. In addition, this

robot has to tailor its tactics simultaneously under the rules of either game. This double complexity of the external dynamics and reward function sometimes becomes more complex when both the multiple dynamics and multiple reward functions switch implicitly, as in the situation of a real (multi-agent) game of tennis where one player cannot observe the intention of her opponents or her partner. The robot must consider its opponent's and its partner's unobservable behavioral goals (reward function). In this article, we address how an RL agent should be designed to handle such double complexity of dynamics and reward. We have previously proposed modular selection and identification for control (MOSAIC) to cope with nonstationary dynamics where appropriate controllers are selected and learned among many candidates based on the error of its paired dynamics predictor: the forward model. Here we extend this framework for RL and propose MOSAIC-MR architecture. It resembles MOSAIC in spirit and selects and learns an appropriate RL controller based on the RL controller's TD error using the errors of the dynamics (the forward model) and the reward predictors. Furthermore, unlike other MOSAIC variants for RL, RL controllers are not a priori paired with the fixed predictors of dynamics and rewards. The simulation results demonstrate that MOSAIC-MR outperforms other counterparts because of this flexible association ability among RL controllers, forward models, and reward predictors.

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21. Efficient Calculation of the Gauss-Newton Approximation of the Hessian Matrix in Neural Networks

Michael Fairbank, Eduardo Alonso

https://doi.org/10.1162/neco_a_00248 | pp. 607-610

The Levenberg-Marquardt (LM) learning algorithm is a popular algorithm for training neural networks; however, for large neural networks, it becomes prohibitively expensive in terms of running time and memory requirements. The most time-critical step of the algorithm is the calculation of the Gauss-Newton matrix, which is formed by multiplying two large Jacobian matrices together. We propose a method that uses backpropagation to reduce the time of this matrix-matrix multiplication. This reduces the overall asymptotic running time of the LM algorithm by a factor of the order of the number of output nodes in the neural network.

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22. Computational Advantages of Reverberating Loops for Sensorimotor Learning

Kristen Fortney, Douglas Tweed

https://doi.org/10.1162/neco_a_00237 | pp. 611-634

When we learn something new, our brain may store the information in synapses or in reverberating loops of electrical activity, but current theories of motor learning focus almost entirely on the synapses. Here we show that loops could also play a role and would bring advantages: loop-based algorithms can learn complex control tasks faster, with exponentially fewer neurons, and avoid the problem of weight transport. They do all this at a cost: in the presence of long feedback delays, loop algorithms cannot control very fast movements, but in this case, loop and synaptic mechanisms can complement each other-mixed systems quickly learn to make accurate but not very fast motions and then gradually speed up. Loop algorithms explain aspects of consolidation, the role of attention, and the relapses that are sometimes seen after a task has apparently been learned, and they make further predictions.

23. Theoretical and Evolutionary Parameter Tuning of Neural Oscillators with a Double-Chain Structure for Generating Rhythmic Signals

Yuya Hattori, Michiyo Suzuki, Zu Soh, Yasuhiko Kobayashi, Toshio Tsuji

https://doi.org/10.1162/neco_a_00249 | pp. 635-675

A neural oscillator with a double-chain structure is one of the central pattern generator models used to simulate and understand rhythmic movements in living organisms. However, it is difficult to reproduce desired rhythmic signals by tuning an enormous number of parameters of neural oscillators. In this study, we propose an automatic tuning method consisting of two parts. The first involves tuning rules for both the time constants and the amplitude of the oscillatory outputs based on theoretical analyses of the relationship between parameters and outputs of the neural oscillators. The second involves an evolutionary tuning method with a two-step genetic algorithm (GA), consisting of a global GA and a local GA, for tuning parameters such as neural connection weights that have no exact tuning rule. Using numerical experiments, we confirmed that the proposed tuning method could successfully tune all parameters and generate sinusoidal waves. The tuning performance of the proposed method was less affected by factors such as the number of excitatory oscillators or the desired outputs. Furthermore, the proposed method was applied to the parameter-tuning problem of some types of artificial and biological wave reproduction and yielded optimal parameter values that generated complex rhythmic signals in Caenorhabditis elegans without trial and error.

24. Noninvertibility, Chaotic Coding, and Chaotic Multiplexity of Synaptically Modulated Neural Firing

Yoram Baram

https://doi.org/10.1162/neco_a_00239 | pp. 676-699

Widely accepted neural firing and synaptic potentiation rules specify a cross-dependence of the two processes, which, evolving on different timescales, have been separated for analytic purposes, concealing essential dynamics. Here, the morphology of the firing rates process, modulated by synaptic potentiation, is shown to be described by a discrete iteration map in the form of a thresholded polynomial. Given initial synaptic weights, a firing activity is triggered by conductance. Elementary dynamic modes are defined by fixed points, cycles, and saddles of the map, building blocks of the underlying firing code. Showing parameter-dependent multiplicity of real polynomial roots, the map is proved to be noninvertible. The incidence of chaos is then implied by the parameter-dependent existence of snap-back repellers. The highly patterned geometric and statistical structures of the associated chaotic attractors suggest that these attractors are an integral part of the neural code. It further suggests the chaotic attractor as a natural mechanism for statistical encoding and temporal multiplexing of neural information. The analytic findings are supported by simulation.

25. Graph Transduction as a Noncooperative Game

Aykut Erdem, Marcello Pelillo

https://doi.org/10.1162/neco_a_00233 | pp. 700-723

Graph transduction is a popular class of semisupervised learning techniques that aims to estimate a classification function defined over a graph of labeled and unlabeled data points. The general idea is to propagate the provided label information to unlabeled nodes in a consistent way. In contrast to the traditional view, in which the process of label propagation is defined as a graph Laplacian regularization, this article proposes a radically different perspective, based on game-theoretic notions. Within the proposed framework, the transduction problem is formulated in terms of a noncooperative multiplayer game whereby equilibria correspond to consistent labelings of the data. An attractive feature of this formulation is that it is inherently a multiclass approach and imposes no constraint whatsoever on the structure of the pairwise similarity matrix, being able to naturally deal with asymmetric and negative similarities alike. Experiments on a number of real-world problems demonstrate that the proposed approach performs well compared with state-of-the-art algorithms, and it can deal effectively with various types of similarity relations.

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26. Simple Modification of Oja Rule Limits L₁-Norm of Weight Vector and Leads to Sparse Connectivity

V. Aparin

https://doi.org/10.1162/neco_a_00240 | pp. 724-743

This letter describes a simple modification of the Oja learning rule, which asymptotically constrains the L1-norm of an input weight vector instead of the L2-norm as in the original rule. This constraining is local as opposed to commonly used instant normalizations, which require the knowledge of all input weights of a neuron to update each one of them individually. The proposed rule converges to a weight vector that is sparser (has more zero weights) than the vector learned by the original Oja rule with or without the zero bound, which could explain the developmental synaptic pruning.

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27. Diffusive Feedback Influences on Hierarchical Information Processing

Ai Miyamoto, Jun Hasegawa, Meihong Zheng, Osamu Hoshino

https://doi.org/10.1162/neco_a_00251 | pp. 744-770

In visual information processing, feedforward projection from primary to secondary visual cortex (V1-to-V2) is essential for integrating combinations of oriented bars in order to extract angular information embedded within contours that represent the shape of objects. For feedback (V2-to-V1) projection, two distinct types of pathways have been observed: clustered projection and diffused projection. The former innervates V1 domains with a preferred orientation similar to that of V2 cells of origin. In contrast, the latter innervates without such orientation specificity. V2 cells send their axons to V1 domains with both similar and dissimilar orientation preferences. It is speculated that the clustered feedback projection has a role in contour integration. The role of the diffused feedback projection, however, remains to be seen. We simulated a minimal, functional V1-V2 neural network model. The diffused feedback projection contributed to achieving ongoing-spontaneous subthreshold membrane oscillations in V1 cells, thereby reducing the reaction time of V1 cells to a pair of bars that represents specific angular information. Interestingly, the feedback influence took place even before V2 responses, which might stem largely from ongoing-spontaneous signaling from V2. We suggest that the diffusive feedback influence from V2 could act early in V1 responses and accelerate their reaction speed to sensory stimulation in order to rapidly extract angular information.

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28. A General Framework for Dimensionality-Reducing Data Visualization Mapping

Kerstin Bunte, Michael Biehl, Barbara Hammer

https://doi.org/10.1162/neco_a_00250 | pp. 771-804

In recent years, a wealth of dimension-reduction techniques for data visualization and preprocessing has been established. Nonparametric methods require additional effort for out-of-sample extensions, because they provide only a mapping of a given finite set of points. In this letter, we propose a general view on nonparametric dimension reduction based on the concept of cost functions and properties of the data. Based on this general principle, we transfer nonparametric dimension reduction to explicit mappings of the data manifold such that direct out-of-sample extensions become possible. Furthermore, this concept offers the possibility of investigating the generalization ability of data visualization to new data points. We demonstrate the approach based on a simple global linear mapping, as well as prototype-based local linear mappings. In addition, we can bias the functional form according to given auxiliary information. This leads to explicit supervised visualization mappings with discriminative properties comparable to state-of-the-art approaches.

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29. Spiking Neural P Systems with Astrocytes

Linqiang Pan, Jun Wang, Hendrik Jan Hoogeboom

https://doi.org/10.1162/neco_a_00238 | pp. 805-825

In a biological nervous system, astrocytes play an important role in the functioning and interaction of neurons, and astrocytes have excitatory and inhibitory influence on synapses. In this work, with this biological inspiration, a class of computation devices that consist of neurons and astrocytes is introduced, called spiking neural P systems with astrocytes (SNPA systems). The computation power of SNPA systems is investigated. It is proved that SNPA systems with simple neurons (all neurons have the same rule, one per neuron, of a very simple form) are Turing universal in both generative and accepting modes. If a bound is given on the number of spikes present in any neuron along a computation, then the computation power of SNPA systems is diminished. In this case, a characterization of semilinear sets of numbers is obtained.

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30. Integration of Reinforcement Learning and Optimal Decision-Making Theories of the Basal Ganglia

Rafa■ Bogacz, Tobias Larsen

https://doi.org/10.1162/neco_a_00103 | pp. 817-851

This article seeks to integrate two sets of theories describing action selection in the basal ganglia: reinforcement learning theories describing learning which actions to select to maximize reward and decision-making theories proposing that the basal ganglia selects actions on the basis of sensory evidence accumulated in the cortex. In particular, we present a model that integrates the actor-critic model of reinforcement learning and a model assuming that the cortico-basal-ganglia circuit implements a statistically optimal decision-making procedure. The values of cortico-striatal weights required for optimal decision making in our model differ from those provided by standard reinforcement learning models. Nevertheless, we show that an actor-critic model converges to the weights required for optimal decision making when

biologically realistic limits on synaptic weights are introduced. We also describe the model's predictions concerning reaction times and neural responses during learning, and we discuss directions required for further integration of reinforcement learning and optimal decision-making theories.

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31. Learning Intermediate-Level Representations of Form and Motion from Natural Movies

Charles F. Cadieu, Bruno A. Olshausen

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We present a model of intermediate-level visual representation that is based on learning invariances from movies of the natural environment. The model is composed of two stages of processing: an early feature representation layer and a second layer in which invariances are explicitly represented. Invariances are learned as the result of factoring apart the temporally stable and dynamic components embedded in the early feature representation. The structure contained in these components is made explicit in the activities of second-layer units that capture invariances in both form and motion. When trained on natural movies, the first layer produces a factorization, or separation, of image content into a temporally persistent part representing local edge structure and a dynamic part representing local motion structure, consistent with known response properties in early visual cortex (area V1). This factorization linearizes statistical dependencies among the first-layer units, making them learnable by the second layer. The second-layer units are split into two populations according to the factorization in the first layer. The form-selective units receive their input from the temporally persistent part (local edge structure) and after training result in a diverse set of higher-order shape features consisting of extended contours, multiscale edges, textures, and texture boundaries. The motion-selective units receive their input from the dynamic part (local motion structure) and after training result in a representation of image translation over different spatial scales and directions, in addition to more complex deformations. These representations provide a rich description of dynamic natural images and testable hypotheses regarding intermediate-level representation in visual cortex.

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32. Quantifying Neurotransmission Reliability Through Metrics-Based Information Analysis

Romain Brasselet, Roland S. Johansson, Angelo Arleo

https://doi.org/10.1162/neco_a_00099 | pp. 852-881

We set forth an information-theoretical measure to quantify neurotransmission reliability while taking into full account the metrical properties of the spike train space. This parametric information analysis relies on similarity measures induced by the metrical relations between neural responses as spikes flow in. Thus, in order to assess the entropy, the conditional entropy, and the overall information transfer, this method does not require any a priori decoding algorithm to partition the space into equivalence classes. It therefore allows the optimal parameters of a class of distances to be determined with respect to information transmission. To validate the proposed information-theoretical approach, we study precise temporal decoding of human somatosensory signals recorded using microneurography experiments. For this analysis, we employ a similarity measure based on the Victor-Purpura spike train metrics. We show that with appropriate parameters of this distance, the relative spike times of the mechanoreceptors' responses convey enough

information to perform optimal discrimination--defined as maximum metrical information and zero conditional entropy--of 81 distinct stimuli within 40 ms of the first afferent spike. The proposed information-theoretical measure proves to be a suitable generalization of Shannon mutual information in order to consider the metrics of temporal codes explicitly. It allows neurotransmission reliability to be assessed in the presence of large spike train spaces (e.g., neural population codes) with high temporal precision.

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33. Decorrelation of Spiking Variability and Improved Information Transfer Through Feedforward Divisive Normalization

Bryan Tripp

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Response variability is often positively correlated in pairs of similarly tuned neurons in the visual cortex. Many authors have considered correlated variability to prevent postsynaptic neurons from averaging across large groups of inputs to obtain reliable stimulus estimates. However, a simple average of variability ignores nonlinearities in cortical signal integration. This study shows that feedforward divisive normalization of a neuron's inputs effectively decorrelates their variability. Furthermore, we show that optimal linear estimates of a stimulus parameter that are based on normalized inputs are more accurate than those based on nonnormalized inputs, due partly to reduced correlations, and that these estimates improve with increasing population size up to several thousand neurons. This suggests that neurons may possess a simple mechanism for substantially decorrelating noise in their inputs. Further work is needed to reconcile this conclusion with past evidence that correlated noise impairs visual perception.

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34. Enhanced Stimulus Encoding Capabilities with Spectral Selectivity in Inhibitory Circuits by STDP

Antoine Coulon, Guillaume Beslon, Hédi Soula

https://doi.org/10.1162/neco_a_00100 | pp. 882-908

The ability to encode and transmit a signal is an essential property that must demonstrate many neuronal circuits in sensory areas in addition to any processing they may provide. It is known that an appropriate level of lateral inhibition, as observed in these areas, can significantly improve the encoding ability of a population of neurons. We show here a homeostatic mechanism by which a spike-timing-dependent plasticity (STDP) rule with a symmetric timing window (swSTDP) spontaneously drives the inhibitory coupling to a level that ensures accurate encoding in response to input signals within a certain frequency range. Interpreting these results mathematically, we find that this coupling level depends on the overlap of spectral information between stimulus and STDP window function. Generalization to arbitrary swSTDP and arbitrary stimuli reveals that the signals for which this improvement of encoding takes place can be finely selected on spectral criteria. We finally show that this spectral overlap principle holds for a variety of neuron types and network characteristics. The highly tunable frequency-power domain of efficiency of this mechanism, together with its ability to operate in very various neuronal contexts, suggest that it may be at work in most sensory areas.

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35. Optimal Sequential Detection of Stimuli from Multiunit Recordings Taken in Densely Populated Brain Regions

Nir Nossenson, Hagit Messer

https://doi.org/10.1162/neco_a_00257 | pp. 895-938

We address the problem of detecting the presence of a recurring stimulus by monitoring the voltage on a multiunit electrode located in a brain region densely populated by stimulus reactive neurons. Published experimental results suggest that under these conditions, when a stimulus is present, the measurements are gaussian with typical second-order statistics. In this letter we systematically derive a generic, optimal detector for the presence of a stimulus in these conditions and describe its implementation. The optimality of the proposed detector is in the sense that it maximizes the life span (or time to injury) of the subject. In addition, we construct a model for the acquired multiunit signal drawing on basic assumptions regarding the nature of a single neuron, which explains the second-order statistics of the raw electrode voltage measurements that are high-pass-filtered above 300 Hz. The operation of the optimal detector and that of a simpler suboptimal detection scheme is demonstrated by simulations and on real electrophysiological data.

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36. An Infomax Algorithm Can Perform Both Familiarity Discrimination and Feature Extraction in a Single Network

Andrew Lulham, Rafa■ Bogacz, Simon Vogt, Malcolm W. Brown

https://doi.org/10.1162/neco_a_00097 | pp. 909-926

Psychological experiments have shown that the capacity of the brain for discriminating visual stimuli as novel or familiar is almost limitless. Neurobiological studies have established that the perirhinal cortex is critically involved in both familiarity discrimination and feature extraction. However, opinion is divided as to whether these two processes are performed by the same neurons. Previously proposed models have been unable to simultaneously extract features and discriminate familiarity for large numbers of stimuli. We show that a well-known model of visual feature extraction, Infomax, can simultaneously perform familiarity discrimination and feature extraction efficiently. This model has a significantly larger capacity than previously proposed models combining these two processes, particularly when correlation exists between inputs, as is the case in the perirhinal cortex. Furthermore, we show that once the model fully extracts features, its ability to perform familiarity discrimination increases markedly.

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37. Synaptic Scaling Stabilizes Persistent Activity Driven by Asynchronous Neurotransmitter Release

Vladislav Volman, Richard C. Gerkin

https://doi.org/10.1162/neco_a_00098 | pp. 927-957

Small networks of cultured hippocampal neurons respond to transient stimulation with rhythmic network activity (reverberation) that persists for several seconds, constituting an in vitro model of synchrony, working memory, and seizure. This mode of activity has been shown theoretically and experimentally to depend on asynchronous neurotransmitter release (an essential feature of the developing hippocampus) and is supported by a variety of developing neuronal networks despite variability in the size of populations (10-200 neurons) and in patterns of synaptic connectivity. It has previously been reported in computational models

that "small-world" connection topology is ideal for the propagation of similar modes of network activity, although this has been shown only for neurons utilizing synchronous (phasic) synaptic transmission. We investigated how topological constraints on synaptic connectivity could shape the stability of reverberations in small networks that also use asynchronous synaptic transmission. We found that reverberation duration in such networks was resistant to changes in topology and scaled poorly with network size. However, normalization of synaptic drive, by reducing the variance of synaptic input across neurons, stabilized reverberation in such networks. Our results thus suggest that the stability of both normal and pathological states in developing networks might be shaped by variance-normalizing constraints on synaptic drive. We offer an experimental prediction for the consequences of such regulation on the behavior of small networks.

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38. Neuronal Responses Below Firing Threshold for Subthreshold Cross-Modal Enhancement

Osamu Hoshino

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Multisensory integration (such as somatosensation-vision, gustation-olfaction) could occur even between subthreshold stimuli that in isolation do not reach perceptual awareness. For example, when a somatosensory (subthreshold) stimulus is delivered within a close spatiotemporal congruency, a visual (subthreshold) stimulus evokes a visual percept. Cross-modal enhancement of visual perception is maximal when the somatosensory stimulation precedes the visual one by tens of milliseconds. This rapid modulatory response would not be consistent with a top-down mechanism acting through higher-order multimodal cortical areas, but rather a direct interaction between lower-order unimodal areas. To elucidate the neuronal mechanisms of subthreshold cross-modal enhancement, we simulated a neural network model. In the model, lower unimodal (X, Y) and higher multimodal (M) networks are reciprocally connected by bottom-up and top-down axonal projections. The lower networks are laterally connected with each other. A pair of stimuli was presented to the lower networks, whose respective intensities were too weak to induce salient neuronal activity (population response) when presented alone. Neurons of the Y network were slightly depolarized below firing threshold when a cross-modal stimulus was presented alone to the X network. This allowed the Y network to make a rapid (within tens of milliseconds) population response when presented with a subsequent congruent stimulus. The reaction speed of the Y network was accelerated, provided that the top-down projections were strengthened. We suggest that a subthreshold (nonpopulation) response to a cross-modal stimulus, acting through interaction between lower (primary unisensory) areas, may be essential for a rapid suprathreshold (population) response to a congruent stimulus that follows. Top-down influences on cross-modal enhancement may be faster than expected, accelerating reaction speed to input, in which ongoing-spontaneous subthreshold excitation of lower-order unimodal cells by higher-order multimodal cells may play an active role.

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39. Regularized Variational Bayesian Learning of Echo State Networks with Delay&Sum Readout

Dmitriy Shutin, Christoph Zechner, Sanjeev R. Kulkarni, H. Vincent Poor

https://doi.org/10.1162/neco_a_00253 | pp. 967-995

In this work, a variational Bayesian framework for efficient training of echo state networks (ESNs) with automatic regularization and delay&sum (D&S) readout adaptation is proposed. The algorithm uses a classical batch learning of ESNs. By treating the network echo states as fixed basis functions parameterized with delay parameters, we propose a variational Bayesian ESN training scheme. The variational approach allows for a seamless combination of sparse Bayesian learning ideas and a variational Bayesian space-alternating generalized expectation-maximization (VB-SAGE) algorithm for estimating parameters of superimposed signals. While the former method realizes automatic regularization of ESNs, which also determines which echo states and input signals are relevant for "explaining" the desired signal, the latter method provides a basis for joint estimation of D&S readout parameters. The proposed training algorithm can naturally be extended to ESNs with fixed filter neurons. It also generalizes the recently proposed expectation-maximization-based D&S readout adaptation method. The proposed algorithm was tested on synthetic data prediction tasks as well as on dynamic handwritten character recognition.

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40. Molecular Diffusion Model of Neurotransmitter Homeostasis Around Synapses Supporting Gradients

Ashwin Mohan, Sandeep Pendyam, Peter W. Kalivas, Satish S. Nair

https://doi.org/10.1162/neco_a_00101 | pp. 984-1014

Neurotransmitter homeostasis in and around a synapse involves complex random processes such as diffusion, molecular binding, and uptake by glial transporters. A three-dimensional stochastic diffusion model of a synapse was developed to provide molecular-level details of neurotransmitter homeostasis not predicted by alternative models based on continuum approaches. The development was illustrated through an example case cortico-accumbens synapse that successfully integrated neuroadaptations observed after chronic cocaine. By incorporating cystine-glutamate exchanger as a nonsynaptic release site for glutamate, the stochastic model was used to quantify the relative contributions of synaptic and nonsynaptic sources to extracellular concentration and to estimate molecular influx rates into the perisynapse. A perturbation analysis showed that among the parameters considered, variation in surface density of glial transporters had the largest effect on glutamate concentrations. The stochastic diffusion model of the example synapse was further generalized to characterize glial morphology by studying the role of diffusion path length in supporting neurotransmitter gradients and isolating the synapse. For the same set of parameters, diffusion path length was found to be proportional to the gradient supported.

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41. Effects of Multiplicative Power Law Neural Noise in Visual Information Processing

José M. Medina

https://doi.org/10.1162/neco_a_00102 | pp. 1015-1046

The human visual system is intrinsically noisy. The benefits of internal noise as part of visual code are controversial. Here the information-theoretic properties of multiplicative (i.e. signal-dependent) neural noise are investigated. A quasi-linear communication channel model is presented. The model shows that multiplicative power law neural noise promotes the minimum information transfer after efficient coding. It is demonstrated that Weber's law and the human contrast sensitivity function arise on the basis of minimum transfer of information and power law neural noise. The implications of minimum information transfer in

self-organized neural networks and weakly coupled neurons are discussed.

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42. Rewiring-Induced Chaos in Pulse-Coupled Neural Networks

Takashi Kanamaru, Kazuyuki Aihara

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The dependence of the dynamics of pulse-coupled neural networks on random rewiring of excitatory and inhibitory connections is examined. When both excitatory and inhibitory connections are rewired, periodic synchronization emerges with a Hopf-like bifurcation and a subsequent period-doubling bifurcation; chaotic synchronization is also observed. When only excitatory connections are rewired, periodic synchronization emerges with a saddle node-like bifurcation, and chaotic synchronization is also observed. This result suggests that randomness in the system does not necessarily contaminate the system, and sometimes it even introduces rich dynamics to the system such as chaos.

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43. Nondegenerate Piecewise Linear Systems: A Finite Newton Algorithm and Applications in Machine Learning

Xiao–Tong Yuan, Shuicheng Yan

https://doi.org/10.1162/neco_a_00241 | pp. 1047-1084

We investigate Newton-type optimization methods for solving piecewise linear systems (PLSs) with nondegenerate coefficient matrix. Such systems arise, for example, from the numerical solution of linear complementarity problem, which is useful to model several learning and optimization problems. In this letter, we propose an effective damped Newton method, PLS-DN, to find the exact (up to machine precision) solution of nondegenerate PLSs. PLS-DN exhibits provable semiiterative property, that is, the algorithm converges globally to the exact solution in a finite number of iterations. The rate of convergence is shown to be at least linear before termination. We emphasize the applications of our method in modeling, from a novel perspective of PLSs, some statistical learning problems such as box-constrained least squares, elitist Lasso (Kowalski & Torreesani, 2008), and support vector machines (Cortes & Vapnik, 1995). Numerical results on synthetic and benchmark data sets are presented to demonstrate the effectiveness and efficiency of PLS-DN on these problems.

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44. Expectation Propagation with Factorizing Distributions: A Gaussian Approximation and Performance Results for Simple Models

Fabiano L. Ribeiro, Manfred Opper

https://doi.org/10.1162/neco_a_00104 | pp. 1047-1069

We discuss the expectation propagation (EP) algorithm for approximate Bayesian inference using a factorizing posterior approximation. For neural network models, we use a central limit theorem argument to make EP tractable when the number of parameters is large. For two types of models, we show that EP can achieve optimal generalization performance when data are drawn from a simple distribution.

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45. Hidden Markov Models for the Stimulus-Response Relationships of Multistate Neural Systems

G. Sean Escola, Alfredo Fontanini, Donald B. Katz, Liam Paninski

https://doi.org/10.1162/neco_a_00118 | pp. 1071-1132

Given recent experimental results suggesting that neural circuits may evolve through multiple firing states, we develop a framework for estimating state-dependent neural response properties from spike train data. We modify the traditional hidden Markov model (HMM) framework to incorporate stimulus-driven, non-Poisson point-process observations. For maximal flexibility, we allow external, time-varying stimuli and the neurons' own spike histories to drive both the spiking behavior in each state and the transitioning behavior between states. We employ an appropriately modified expectation-maximization algorithm to estimate the model parameters. The expectation step is solved by the standard forward-backward algorithm for HMMs. The maximization step reduces to a set of separable concave optimization problems if the model is restricted slightly. We first test our algorithm on simulated data and are able to fully recover the parameters used to generate the data and accurately recapitulate the sequence of hidden states. We then apply our algorithm to a recently published data set in which the observed neuronal ensembles displayed multistate behavior and show that inclusion of spike history information significantly improves the fit of the model. Additionally, we show that a simple reformulation of the state space of the underlying Markov chain allows us to implement a hybrid half-multistate, half-histogram model that may be more appropriate for capturing the complexity of certain data sets than either a simple HMM or a simple peristimulus time histogram model alone.

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46. Accelerated Multiplicative Updates and Hierarchical ALS Algorithms for Nonnegative Matrix Factorization

Nicolas Gillis, François Glineur

https://doi.org/10.1162/neco_a_00256 | pp. 1085-1105

Nonnegative matrix factorization (NMF) is a data analysis technique used in a great variety of applications such as text mining, image processing, hyperspectral data analysis, computational biology, and clustering. In this letter, we consider two well-known algorithms designed to solve NMF problems: the multiplicative updates of Lee and Seung and the hierarchical alternating least squares of Cichocki et al. We propose a simple way to significantly accelerate these schemes, based on a careful analysis of the computational cost needed at each iteration, while preserving their convergence properties. This acceleration technique can also be applied to other algorithms, which we illustrate on the projected gradient method of Lin. The efficiency of the accelerated algorithms is empirically demonstrated on image and text data sets and compares favorably with a state-of-the-art alternating nonnegative least squares algorithm.

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47. Revisiting Tests for Neglected Nonlinearity Using Artificial Neural Networks

Jin Seo Cho, Isao Ishida, Halbert White

https://doi.org/10.1162/neco_a_00117 | pp. 1133-1186

Tests for regression neglected nonlinearity based on artificial neural networks (ANNs) have so far been studied by separately analyzing the two ways in which the null of regression linearity can hold. This implies that the asymptotic behavior of general ANN-based tests for neglected nonlinearity is still an open question. Here we analyze a convenient ANN-based quasi-likelihood ratio statistic for testing neglected nonlinearity, paying careful attention to both components of the null. We derive the asymptotic null distribution under each component separately and analyze their interaction. Somewhat remarkably, it turns out that the previously known asymptotic null distribution for the type 1 case still applies, but under somewhat stronger conditions than previously recognized. We present Monte Carlo experiments corroborating our theoretical results and showing that standard methods can yield misleading inference when our new, stronger regularity conditions are violated.

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48. On the Performance of Voltage Stepping for the Simulation of Adaptive, Nonlinear Integrate-and-Fire Neuronal Networks

Mohamed Ghaith Kaabi, Arnaud Tonnelier, Dominique Martinez

https://doi.org/10.1162/neco_a_00112 | pp. 1187-1204

In traditional event-driven strategies, spike timings are analytically given or calculated with arbitrary precision (up to machine precision). Exact computation is possible only for simplified neuron models, mainly the leaky integrate-and-fire model. In a recent paper, Zheng, Tonnelier, and Martinez (2009) introduced an approximate event-driven strategy, named voltage stepping, that allows the generic simulation of nonlinear spiking neurons. Promising results were achieved in the simulation of single quadratic integrate-and-fire neurons. Here, we assess the performance of voltage stepping in network simulations by considering more complex neurons (quadratic integrate-and-fire neurons with adaptation) coupled with multiple synapses. To handle the discrete nature of synaptic interactions, we recast voltage stepping in a general framework, the discrete event system specification. The efficiency of the method is assessed through simulations and comparisons with a modified time-stepping scheme of the Runge-Kutta type. We demonstrated numerically that the original order of voltage stepping is preserved when simulating connected spiking neurons, independent of the network activity and connectivity.

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49. Detection of Hidden Structures in Nonstationary Spike Trains

Ken Takiyama, Masato Okada

https://doi.org/10.1162/neco_a_00109 | pp. 1205-1233

We propose an algorithm for simultaneously estimating state transitions among neural states and nonstationary firing rates using a switching state-space model (SSSM). This algorithm enables us to detect state transitions on the basis of not only discontinuous changes in mean firing rates but also discontinuous changes in the temporal profiles of firing rates (e.g., temporal correlation). We construct estimation and learning algorithms for a nongaussian SSSM, whose nongaussian property is caused by binary spike events. Local variational methods can transform the binary observation process into a quadratic form. The transformed observation process enables us to construct a variational Bayes algorithm that can determine the number of neural states based on automatic relevance determination. Additionally, our algorithm can estimate model parameters from single-trial data using a priori knowledge about state transitions and firing rates. Synthetic data analysis reveals that our algorithm has higher performance for estimating

nonstationary firing rates than previous methods. The analysis also confirms that our algorithm can detect state transitions on the basis of discontinuous changes in temporal correlation, which are transitions that previous hidden Markov models could not detect. We also analyze neural data recorded from the medial temporal area. The statistically detected neural states probably coincide with transient and sustained states that have been detected heuristically. Estimated parameters suggest that our algorithm detects the state transitions on the basis of discontinuous changes in the temporal correlation of firing rates. These results suggest that our algorithm is advantageous in real-data analysis.

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50. The Dynamics of Integrate-and-Fire: Mean Versus Variance Modulations and Dependence on Baseline Parameters

Joanna Pressley, Todd W. Troyer

https://doi.org/10.1162/neco_a_00114 | pp. 1234-1247

The leaky integrate-and-fire (LIF) is the simplest neuron model that captures the essential properties of neuronal signaling. Yet common intuitions are inadequate to explain basic properties of LIF responses to sinusoidal modulations of the input. Here we examine responses to low and moderate frequency modulations of both the mean and variance of the input current and quantify how these responses depend on baseline parameters. Across parameters, responses to modulations in the mean current are low pass, approaching zero in the limit of high frequencies. For very low baseline firing rates, the response cutoff frequency matches that expected from membrane integration. However, the cutoff shows a rapid, supralinear increase with firing rate, with a steeper increase in the case of lower noise. For modulations of the input variance, the gain at high frequency remains finite. Here, we show that the low-frequency responses depend strongly on baseline parameters and derive an analytic condition specifying the parameters at which responses switch from being dominated by low versus high frequencies. Additionally, we show that the resonant responses for variance modulations have properties not expected for common oscillatory resonances: they peak at frequencies higher than the baseline firing rate and persist when oscillatory spiking is disrupted by high noise. Finally, the responses to mean and variance modulations are shown to have a complementary dependence on baseline parameters at higher frequencies, resulting in responses to modulations of Poisson input rates that are independent of baseline input statistics.

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51. Traveling Bumps and Their Collisions in a Two-Dimensional Neural Field

Yao Lu, Yuzuru Sato, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco_a_00111 | pp. 1248-1260

A neural field is a continuous version of a neural network model accounting for dynamical pattern forming from populational firing activities in neural tissues. These patterns include standing bumps, moving bumps, traveling waves, target waves, breathers, and spiral waves, many of them observed in various brain areas. They can be categorized into two types: a wave-like activity spreading over the field and a particle-like localized activity. We show through numerical experiments that localized traveling excitation patterns (traveling bumps), which behave like particles, exist in a two-dimensional neural field with excitation and inhibition mechanisms. The traveling bumps do not require any geometric restriction (boundary) to prevent them from propagating away, a fact that might shed light on how neurons in the brain are functionally organized. Collisions of traveling bumps exhibit rich phenomena; they might reveal the manner of

information processing in the cortex and be useful in various applications. The trajectories of traveling bumps can be controlled by external inputs.

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52. Mechanisms That Modulate the Transfer of Spiking Correlations

Robert Rosenbaum, Krešimir Josi■■

https://doi.org/10.1162/neco_a_00116 | pp. 1261-1305

Correlations between neuronal spike trains affect network dynamics and population coding. Overlapping afferent populations and correlations between presynaptic spike trains introduce correlations between the inputs to downstream cells. To understand network activity and population coding, it is therefore important to understand how these input correlations are transferred to output correlations.Recent studies have addressed this question in the limit of many inputs with infinitesimal postsynaptic response amplitudes, where the total input can be approximated by gaussian noise. In contrast, we address the problem of correlation transfer by representing input spike trains as point processes, with each input spike eliciting a finite postsynaptic response. This approach allows us to naturally model synaptic noise and recurrent coupling and to treat excitatory and inhibitory inputs separately.We derive several new results that provide intuitive insights into the fundamental mechanisms that modulate the transfer of spiking correlations.

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53. Metacognitive Learning in a Fully Complex-Valued Radial Basis Function Neural Network

R. Savitha, S. Suresh, N. Sundararajan

https://doi.org/10.1162/neco_a_00254 | pp. 1297-1328

Recent studies on human learning reveal that self-regulated learning in a metacognitive framework is the best strategy for efficient learning. As the machine learning algorithms are inspired by the principles of human learning, one needs to incorporate the concept of metacognition to develop efficient machine learning algorithms. In this letter we present a metacognitive learning framework that controls the learning process of a fully complex-valued radial basis function network and is referred to as a metacognitive fully complex-valued radial basis function (Mc-FCRBF) network. Mc-FCRBF has two components: a cognitive component containing the FC-RBF network and a metacognitive component, which regulates the learning process of FC-RBF. In every epoch, when a sample is presented to Mc-FCRBF, the metacognitive component decides what to learn, when to learn, and how to learn based on the knowledge acquired by the FC-RBF network and the new information contained in the sample. The Mc-FCRBF learning algorithm is described in detail, and both its approximation and classification abilities are evaluated using a set of benchmark and practical problems. Performance results indicate the superior approximation and classification performance of Mc-FCRBF compared to existing methods in the literature.

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54. Refinements of Universal Approximation Results for Deep Belief Networks and Restricted Boltzmann Machines

Guido Montúfar, Nihat Ay

https://doi.org/10.1162/neco_a_00113 | pp. 1306-1319

We improve recently published results about resources of restricted Boltzmann machines (RBM) and deep belief networks (DBN)required to make them universal approximators. We show that any distribution pon the set {0,1}(n) of binary vectors of length n can be arbitrarily well approximated by an RBM with k-1 hidden units, where k is the minimal number of pairs of binary vectors differing in only one entry such that their union contains the support set of p. In important cases this number is half the cardinality of the support set of p (given in Le Roux & Bengio, 2008). We construct a DBN with 2n/ 2(n-b) , b ~ log n, hidden layers of width n that is capable of approximating any distribution on {0,1}(n) arbitrarily well. This confirms a conjecture presented in Le Roux and Bengio (2010).

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55. Time-Free Spiking Neural P Systems

Linqiang Pan, Xiangxiang Zeng, Xingyi Zhang

https://doi.org/10.1162/neco_a_00115 | pp. 1320-1342

Different biological processes take different times to be completed, which can also be influenced by many environmental factors. In this work, a realistic definition of nonsynchronized spiking neural P systems (SN P systems, for short) is considered: during the work of an SN P system, the execution times of spiking rules cannot be known exactly (i.e., they are arbitrary). In order to establish robust systems against the environmental factors, a special class of SN P systems, called time-free SN P systems, is introduced, which always produce the same computation result independent of the execution times of the rules. The universality of time-free SN P systems is investigated. It is proved that these P systems with extended rules (several spikes can be produced by a rule) are equivalent to register machines. However, if the number of spikes present in the system is bounded, then the power of time-free SN P systems falls, and in this case, a characterization of semilinear sets of natural numbers is obtained.

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56. Divergence-Based Vector Quantization

Thomas Villmann, Sven Haase

https://doi.org/10.1162/neco_a_00110 | pp. 1343-1392

Supervised and unsupervised vector quantization methods for classification and clustering traditionally use dissimilarities, frequently taken as Euclidean distances. In this article, we investigate the applicability of divergences instead, focusing on online learning. We deduce the mathematical fundamentals for its utilization in gradient-based online vector quantization algorithms. It bears on the generalized derivatives of the divergences known as Fréchet derivatives in functional analysis, which reduces in finite-dimensional problems to partial derivatives in a natural way. We demonstrate the application of this methodology for widely applied supervised and unsupervised online vector quantization schemes, including self-organizing maps, neural gas, and learning vector quantization. Additionally, principles for hyperparameter optimization and relevance learning for parameterized divergences in the case of supervised vector quantization are given to achieve improved classification accuracy.

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57. Neural Associative Memory with Optimal Bayesian Learning

Andreas Knoblauch

https://doi.org/10.1162/neco_a_00127 | pp. 1393-1451

Neural associative memories are perceptron-like single-layer networks with fast synaptic learning typically storing discrete associations between pairs of neural activity patterns. Previous work optimized the memory capacity for various models of synaptic learning: linear Hopfield-type rules, the Willshaw model employing binary synapses, or the BCPNN rule of Lansner and Ekeberg, for example. Here I show that all of these previous models are limit cases of a general optimal model where synaptic learning is determined by probabilistic Bayesian considerations. Asymptotically, for large networks and very sparse neuron activity, the Bayesian model becomes identical to an inhibitory implementation of the Willshaw and BCPNN-type models. For less sparse patterns, the Bayesian model becomes identical to Hopfield-type networks employing the covariance rule. For intermediate sparseness or finite networks, the optimal Bayesian learning rule differs from the previous models and can significantly improve memory performance. I also provide a unified analytical framework to determine memory capacity at a given output noise level that links approaches based on mutual information, Hamming distance, and signal-to-noise ratio.

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58. Applying the Multivariate Time-Rescaling Theorem to Neural Population Models

Felipe Gerhard, Robert Haslinger, Gordon Pipa

https://doi.org/10.1162/neco_a_00126 | pp. 1452-1483

Statistical models of neural activity are integral to modern neuroscience. Recently interest has grown in modeling the spiking activity of populations of simultaneously recorded neurons to study the effects of correlations and functional connectivity on neural information processing. However, any statistical model must be validated by an appropriate goodness-of-fit test. Kolmogorov-Smirnov tests based on the time-rescaling theorem have proven to be useful for evaluating point-process-based statistical models of single-neuron spike trains. Here we discuss the extension of the time-rescaling theorem to the multivariate (neural population) case. We show that even in the presence of strong correlations between spike trains, models that neglect couplings between neurons can be erroneously passed by the univariate time-rescaling test. We present the multivariate version of the time-rescaling theorem and provide a practical step-by-step procedure for applying it to testing the sufficiency of neural population models. Using several simple analytically tractable models and more complex simulated and real data sets, we demonstrate that important features of the population activity can be detected only using the multivariate extension of the test.

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59. Insights from a Simple Expression for Linear Fisher Information in a Recurrently Connected Population of Spiking Neurons

Jeffrey M. Beck, Vikranth R. Bejjanki, Alexandre Pouget

https://doi.org/10.1162/neco_a_00125 | pp. 1484-1502

A simple expression for a lower bound of Fisher information is derived for a network of recurrently connected spiking neurons that have been driven to a noise-perturbed steady state. We call this lower bound linear Fisher information, as it corresponds to the Fisher information that can be recovered by a locally optimal linear estimator. Unlike recent similar calculations, the approach used here includes the effects of nonlinear gain functions and correlated input noise and yields a surprisingly simple and intuitive expression that offers

substantial insight into the sources of information degradation across successive layers of a neural network. Here, this expression is used to (1) compute the optimal (i.e., information-maximizing) firing rate of a neuron, (2) demonstrate why sharpening tuning curves by either thresholding or the action of recurrent connectivity is generally a bad idea, (3) show how a single cortical expansion is sufficient to instantiate a redundant population code that can propagate across multiple cortical layers with minimal information loss, and (4) show that optimal recurrent connectivity strongly depends on the covariance structure of the inputs to the network.

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60. Vectorized Algorithms for Spiking Neural Network Simulation

Romain Brette, Dan F. M. Goodman

https://doi.org/10.1162/neco_a_00123 | pp. 1503-1535

High-level languages (Matlab, Python) are popular in neuroscience because they are flexible and accelerate development. However, for simulating spiking neural networks, the cost of interpretation is a bottleneck. We describe a set of algorithms to simulate large spiking neural networks efficiently with high-level languages using vector-based operations. These algorithms constitute the core of Brian, a spiking neural network simulator written in the Python language. Vectorized simulation makes it possible to combine the flexibility of high-level languages with the computational efficiency usually associated with compiled languages.

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61. Multiplicative Gain Modulation Arises Through Unsupervised Learning in a Predictive Coding Model of Cortical Function

Kris De Meyer, Michael Spratling

https://doi.org/10.1162/neco_a_00130 | pp. 1536-1567

The combination of two or more population-coded signals in a neural model of predictive coding can give rise to multiplicative gain modulation in the response properties of individual neurons. Synaptic weights generating these multiplicative response properties can be learned using an unsupervised, Hebbian learning rule. The behavior of the model is compared to empirical data on gaze-dependent gain modulation of cortical cells and found to be in good agreement with a range of physiological observations. Furthermore, it is demonstrated that the model can learn to represent a set of basis functions. This letter thus connects an often-observed neurophysiological phenomenon and important neurocomputational principle (gain modulation) with an influential theory of brain operation (predictive coding).

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62. Projective Clustering Using Neural Networks with Adaptive Delay and Signal Transmission Loss

Jianhong Wu, Hossein ZivariPiran, John D. Hunter, John Milton

https://doi.org/10.1162/neco_a_00124 | pp. 1568-1604

We develop a new neural network architecture for projective clustering of data sets that incorporates adaptive transmission delays and signal transmission information loss. The resultant selective output signaling mechanism does not require the addition of multiple hidden layers but instead is based on the assumption that the signal transmission velocity between input processing neurons and clustering neurons

is proportional to the similarity between the input pattern and the feature vector (the top-down weights) of the clustering neuron. The mathematical model governing the evolution of the signal transmission delay, the short-term memory traces, and the long-term memory traces represents a new class of large-scale delay differential equations where the evolution of the delay is described by a nonlinear differential equation involving the similarity measure already noted. We give a complete description of the computational performance of the network for a wide range of parameter values.

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63. Regularized Parameter Estimation in High-Dimensional Gaussian Mixture Models

Lingyan Ruan, Ming Yuan, Hui Zou

https://doi.org/10.1162/neco_a_00128 | pp. 1605-1622

Finite gaussian mixture models are widely used in statistics thanks to their great flexibility. However, parameter estimation for gaussian mixture models with high dimensionality can be challenging because of the large number of parameters that need to be estimated. In this letter, we propose a penalized likelihood estimator to address this difficulty. The [Formula: see text]-type penalty we impose on the inverse covariance matrices encourages sparsity on its entries and therefore helps to reduce the effective dimensionality of the problem. We show that the proposed estimate can be efficiently computed using an expectation-maximization algorithm. To illustrate the practical merits of the proposed method, we consider its applications in model-based clustering and mixture discriminant analysis. Numerical experiments with both simulated and real data show that the new method is a valuable tool for high-dimensional data analysis.

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64. An Estimation of Generalized Bradley-Terry Models Based on the em Algorithm

Yu Fujimoto, Hideitsu Hino, Noboru Murata

https://doi.org/10.1162/neco_a_00129 | pp. 1623-1659

The Bradley-Terry model is a statistical representation for one's preference or ranking data by using pairwise comparison results of items. For estimation of the model, several methods based on the sum of weighted Kullback-Leibler divergences have been proposed from various contexts. The purpose of this letter is to interpret an estimation mechanism of the Bradley-Terry model from the viewpoint of flatness, a fundamental notion used in information geometry. Based on this point of view, a new estimation method is proposed on a framework of the em algorithm. The proposed method is different in its objective function from that of conventional methods, especially in treating unobserved comparisons, and it is consistently interpreted in a probability simplex. An estimation method with weight adaptation is also proposed from a viewpoint of the sensitivity. Experimental results show that the proposed method works appropriately, and weight adaptation improves accuracy of the estimate.

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65. A Connection Between Score Matching and Denoising Autoencoders

Pascal Vincent

https://doi.org/10.1162/neco_a_00142 | pp. 1661-1674

Denoising autoencoders have been previously shown to be competitive alternatives to restricted Boltzmann machines for unsupervised pretraining of each layer of a deep architecture. We show that a simple denoising autoencoder training criterion is equivalent to matching the score (with respect to the data) of a specific energy-based model to that of a nonparametric Parzen density estimator of the data. This yields several useful insights. It defines a proper probabilistic model for the denoising autoencoder technique, which makes it in principle possible to sample from them or rank examples by their energy. It suggests a different way to apply score matching that is related to learning to denoise and does not require computing second derivatives. It justifies the use of tied weights between the encoder and decoder and suggests ways to extend the success of denoising autoencoders to a larger family of energy-based models.

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66. Spiking Neurons and the First Passage Problem

Lawrence Sirovich, Bruce W. Knight

https://doi.org/10.1162/neco_a_00139 | pp. 1675-1703

We derive a model of a neuron's interspike interval probability density through analysis of the first passage problem. The fit of our expression to retinal ganglion cell laboratory data extracts three physiologically relevant parameters, with which our model yields input-output features that conform to laboratory results. Preliminary analysis suggests that under common circumstances, local circuitry readjusts these parameters with changes in firing rate and so endeavors to faithfully replicate an input signal. Further results suggest that the so-called principle of sloppy workmanship also plays a role in evolution's choice of these parameters.

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67. On the Simulation of Nonlinear Bidimensional Spiking Neuron Models

Jonathan Touboul

https://doi.org/10.1162/neco_a_00141 | pp. 1704-1742

Bidimensional spiking models are garnering a lot of attention for their simplicity and their ability to reproduce various spiking patterns of cortical neurons and are used particularly for large network simulations. These models describe the dynamics of the membrane potential by a nonlinear differential equation that blows up in finite time, coupled to a second equation for adaptation. Spikes are emitted when the membrane potential blows up or reaches a cutoff θ. The precise simulation of the spike times and of the adaptation variable is critical, for it governs the spike pattern produced and is hard to compute accurately because of the exploding nature of the system at the spike times. We thoroughly study the precision of fixed time-step integration schemes for this type of model and demonstrate that these methods produce systematic errors that are unbounded, as the cutoff value is increased, in the evaluation of the two crucial quantities: the spike time and the value of the adaptation variable at this time. Precise evaluation of these quantities therefore involves very small time steps and long simulation times. In order to achieve a fixed absolute precision in a reasonable computational time, we propose here a new algorithm to simulate these systems based on a variable integration step method that either integrates the original ordinary differential equation or the equation of the orbits in the phase plane, and compare this algorithm with fixed time-step Euler scheme and other more accurate simulation algorithms.

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68. How Sample Paths of Leaky Integrate-and-Fire Models Are Influenced by the Presence of a Firing Threshold

Mar■■a Teresa Giraudo, Priscilla E. Greenwood, Laura Sacerdote

https://doi.org/10.1162/neco_a_00143 | pp. 1743-1767

Neural membrane potential data are necessarily conditional on observation being prior to a firing time. In a stochastic leaky integrate-and-fire model, this corresponds to conditioning the process on not crossing a boundary. In the literature, simulation and estimation have almost always been done using unconditioned processes. In this letter, we determine the stochastic differential equations of a diffusion process conditioned to stay below a level S up to a fixed time t(1) and of a diffusion process conditioned to cross the boundary for the first time at t(1). This allows simulation of sample paths and identification of the corresponding mean process. Differences between the mean of free and conditioned processes are illustrated, as well as the role of noise in increasing these differences.

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69. Spike-Timing-Dependent Plasticity and Reliability Optimization: The Role of Neuron Dynamics

Román Rossi-Pool, Germán Mato

https://doi.org/10.1162/neco_a_00140 | pp. 1768-1789

Plastic changes in synaptic efficacy can depend on the time ordering of presynaptic and postsynaptic spikes. This phenomenon is called spike-timing-dependent plasticity (STDP). One of the most striking aspects of this plasticity mechanism is that the STDP windows display a great variety of forms in different parts of the nervous system. We explore this issue from a theoretical point of view. We choose as the optimization principle the minimization of conditional entropy or maximization of reliability in the transmission of information. We apply this principle to two types of postsynaptic dynamics, designated type I and type II. The first is characterized as being an integrator, while the second is a resonator. We find that, depending on the parameters of the models, the optimization principle can give rise to a wide variety of STDP windows, such as antisymmetric Hebbian, predominantly depressing or symmetric with one positive region and two lateral negative regions. We can relate each of these forms to the dynamical behavior of the different models. We also propose experimental tests to assess the validity of the optimization principle.

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70. Inhibition in Superior Colliculus Neurons in a Brightness Discrimination Task?

Roger Ratcliff, Yukako T. Hasegawa, Ryohei Hasegawa, Russ Childers, Philip L. Smith, Mark A. Segraves

https://doi.org/10.1162/neco_a_00135 | pp. 1790-1820

Simultaneous recordings were collected from between two and four buildup neurons from the left and right superior colliculi in rhesus monkeys in a simple two-choice brightness discrimination task. The monkeys were required to move their eyes to one of two response targets to indicate their decision. Neurons were identified whose receptive fields were centered on the response targets. The functional role of inhibition was examined by conditionalizing firing rate on a high versus low rate in target neurons 90 ms to 30 ms before the saccade and examining the firing rate in both contralateral and ipsilateral neurons. Two models with racing diffusion processes were fit to the behavioral data, and the same analysis was performed on

simulated paths in the diffusion processes that have been found to represent firing rate. The results produce converging evidence for the lack of a functional role for inhibition between neural populations corresponding to the two decisions.

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71. A Neural Network Model for the Acquisition of a Spatial Body Scheme Through Sensorimotor Interaction

V. Y. Roschin, Alexander Frolov, Yves Burnod, Marc A. Maier

https://doi.org/10.1162/neco_a_00138 | pp. 1821-1834

This letter presents a novel unsupervised sensory matching learning technique for the development of an internal representation of three-dimensional information. The representation is invariant with respect to the sensory modalities involved. Acquisition of the internal representation is demonstrated with a neural network model of a sensorimotor system of a simple model creature, consisting of a tactile-sensitive body and a multiple-degrees-of-freedom arm with proprioceptive sensitivity. Acquisition of the 3D representation as well as a distributed representation of the body scheme, occurs through sensorimotor interactions (i.e., the sensory-motor experience of the creature). Convergence of the learning is demonstrated through computer simulations for the model creature with a 7-DoF arm and a spherical body covered by 20 tactile fields.

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72. A Multifactor Winner-Take-All Dynamics

Junmei Zhu

https://doi.org/10.1162/neco_a_00136 | pp. 1835-1861

Perceptual systems often have to disentangle different factors from mixed observations. If each factor is represented by a set of variables, each standing for a discrete value of the factor, the factor values underlying an observation can be extracted by a winner-take-all (WTA) mechanism over the direct product of the factors. Search in the product space, however, is expensive. It is computationally attractive to work on the marginal factors. In this letter we study the dynamics of a multifactor system modeled by a number of interacting WTA dynamics, one for each factor. We give theoretical results on the stable fixed points of this system and show experimental results on invariant object recognition.

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73. A Finite-Sample, Distribution-Free, Probabilistic Lower Bound on Mutual Information

Nathan D. VanderKraats, Arunava Banerjee

https://doi.org/10.1162/neco_a_00144 | pp. 1862-1898

For any memoryless communication channel with a binary-valued input and a one-dimensional real-valued output, we introduce a probabilistic lower bound on the mutual information given empirical observations on the channel. The bound is built on the Dvoretzky-Kiefer-Wolfowitz inequality and is distribution free. A quadratic time algorithm is described for computing the bound and its corresponding class-conditional distribution functions. We compare our approach to existing techniques and show the superiority of our bound to a method inspired by Fano's inequality where the continuous random variable is discretized.

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74. Determination and the No-Free-Lunch Paradox

Etienne Barnard

https://doi.org/10.1162/neco_a_00137 | pp. 1899-1909

We discuss the no-free-lunch NFL theorem for supervised learning as a logical paradox--that is, as a counterintuitive result that is correctly proven from apparently incontestable assumptions. We show that the uniform prior that is used in the proof of the theorem has a number of unpalatable consequences besides the NFL theorem, and propose a simple definition of determination (by a learning set of given size) that casts additional suspicion on the utility of this assumption for the prior. Whereas others have suggested that the assumptions of the NFL theorem are not practically realistic, we show these assumptions to be at odds with supervised learning in principle. This analysis suggests a route toward the establishment of a more realistic prior probability for use in the extended Bayesian framework.

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75. Distributed Control of Uncertain Systems Using Superpositions of Linear Operators

Terence D. Sanger

https://doi.org/10.1162/neco_a_00151 | pp. 1911-1934

Control in the natural environment is difficult in part because of uncertainty in the effect of actions. Uncertainty can be due to added motor or sensory noise, unmodeled dynamics, or quantization of sensory feedback. Biological systems are faced with further difficulties, since control must be performed by networks of cooperating neurons and neural subsystems. Here, we propose a new mathematical framework for modeling and simulation of distributed control systems operating in an uncertain environment. Stochastic differential operators can be derived from the stochastic differential equation describing a system, and they map the current state density into the differential of the state density. Unlike discrete-time Markov update operators, stochastic differential operators combine linearly for a large class of linear and nonlinear systems, and therefore the combined effects of multiple controllable and uncontrollable subsystems can be predicted. Design using these operators yields systems whose statistical behavior can be specified throughout state-space. The relationship to Bayesian estimation and discrete-time Markov processes is described.

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76. A Simple Derivation of a Bound on the Perceptron Margin Using Singular Value Decomposition

Omri Barak, Mattia Rigotti

https://doi.org/10.1162/neco_a_00152 | pp. 1935-1943

The perceptron is a simple supervised algorithm to train a linear classifier that has been analyzed and used extensively. The classifier separates the data into two groups using a decision hyperplane, with the margin between the data and the hyperplane determining the classifier's ability to generalize and its robustness to input noise. Exact results for the maximal size of the separating margin are known for specific input distributions, and bounds exist for arbitrary distributions, but both rely on lengthy statistical mechanics calculations carried out in the limit of infinite input size. Here we present a short analysis of perceptron classification using singular value decomposition. We provide a simple derivation of a lower bound on the margin and an explicit formula for the perceptron weights that converges to the optimal result for large

separating margins.

77. Firing Variability Is Higher than Deduced from the Empirical Coefficient of Variation

Susanne Ditlevsen, Petr Lánský

https://doi.org/10.1162/neco_a_00157 | pp. 1944-1966

A convenient and often used summary measure to quantify the firing variability in neurons is the coefficient of variation (CV), defined as the standard deviation divided by the mean. It is therefore important to find an estimator that gives reliable results from experimental data, that is, the estimator should be unbiased and have low estimation variance. When the CV is evaluated in the standard way (empirical standard deviation of interspike intervals divided by their average), then the estimator is biased, underestimating the true CV, especially if the distribution of the interspike intervals is positively skewed. Moreover, the estimator has a large variance for commonly used distributions. The aim of this letter is to quantify the bias and propose alternative estimation methods. If the distribution is assumed known or can be determined from data, parametric estimators are proposed, which not only remove the bias but also decrease the estimation errors. If no distribution is assumed and the data are very positively skewed, we propose to correct the standard estimator. When defining the corrected estimator, we simply use that it is more stable to work on the log scale for positively skewed distributions. The estimators are evaluated through simulations and applied to experimental data from olfactory receptor neurons in rats.

78. Online Variational Inference for State-Space Models with Point-Process Observations

Andrew Zammit■Mangion, Ke Yuan, Visakan Kadirkamanathan, Mahesan Niranjan, Guido Sanguinetti

https://doi.org/10.1162/neco_a_00156 | pp. 1967-1999

We present a variational Bayesian (VB) approach for the state and parameter inference of a state-space model with point-process observations, a physiologically plausible model for signal processing of spike data. We also give the derivation of a variational smoother, as well as an efficient online filtering algorithm, which can also be used to track changes in physiological parameters. The methods are assessed on simulated data, and results are compared to expectation-maximization, as well as Monte Carlo estimation techniques, in order to evaluate the accuracy of the proposed approach. The VB filter is further assessed on a data set of taste-response neural cells, showing that the proposed approach can effectively capture dynamical changes in neural responses in real time.

79. Diffusive Information Accumulation by Minimal Recurrent Neural Models of Decision Making

Philip L. Smith, Cameron R. L. McKenzie

https://doi.org/10.1162/neco_a_00150 | pp. 2000-2031

An important class of psychological models of decision making assumes that evidence is accumulated by a diffusion process to a response criterion. These models have successfully accounted for reaction time (RT)

distributions and choice probabilities from a wide variety of experimental tasks. An outstanding theoretical problem is how the integration process that underlies diffusive evidence accumulation can be realized neurally. Wang (2001 , 2002) has suggested that long timescale neural integration may be implemented by persistent activity in reverberation loops. We analyze a simple recurrent decision making architecture and show that it leads to a diffusive accumulation process. The process has the form of a time-inhomogeneous Ornstein-Uhlenbeck velocity process with linearly increasing drift and diffusion coefficients. The resulting model predicts RT distributions and choice probabilities that closely approximate those found in behavioral data.

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80. A New Clustering Approach on the Basis of Dynamical Neural Field

Dequan Jin, Jigen Peng, ■■■ ■■

https://doi.org/10.1162/neco_a_00153 | pp. 2032-2057

In this letter, we present a new hierarchical clustering approach based on the evolutionary process of Amari's dynamical neural field model. Dynamical neural field theory provides a theoretical framework macroscopically describing the activity of neuron ensemble. Based on it, our clustering approach is essentially close to the neurophysiological nature of perception. It is also computationally stable, insensitive to noise, flexible, and tractable for data with complex structure. Some examples are given to show the feasibility.

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81. Quickly Generating Representative Samples from an RBM-Derived Process

Olivier Breuleux, Yoshua Bengio, Pascal Vincent

https://doi.org/10.1162/neco_a_00158 | pp. 2058-2073

Two recently proposed learning algorithms, herding and fast persistent contrastive divergence (FPCD), share the following interesting characteristic: they exploit changes in the model parameters while sampling in order to escape modes and mix better during the sampling process that is part of the learning algorithm. We justify such approaches as ways to escape modes while keeping approximately the same asymptotic distribution of the Markov chain. In that spirit, we extend FPCD using an idea borrowed from Herding in order to obtain a pure sampling algorithm, which we call the rates-FPCD sampler. Interestingly, this sampler can improve the model as we collect more samples, since it optimizes a lower bound on the log likelihood of the training data. We provide empirical evidence that this new algorithm displays substantially better and more robust mixing than Gibbs sampling.

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82. A Regularized Correntropy Framework for Robust Pattern Recognition

Ran He, Wei■Shi Zheng, Bao-Gang Hu, Xiangwei Kong

https://doi.org/10.1162/neco_a_00155 | pp. 2074-2100

This letter proposes a new multiple linear regression model using regularized correntropy for robust pattern recognition. First, we motivate the use of correntropy to improve the robustness of the classical mean square error (MSE) criterion that is sensitive to outliers. Then an l 1 regularization scheme is imposed on the

correntropy to learn robust and sparse representations. Based on the half-quadratic optimization technique, we propose a novel algorithm to solve the nonlinear optimization problem. Second, we develop a new correntropy-based classifier based on the learned regularization scheme for robust object recognition. Extensive experiments over several applications confirm that the correntropy-based l 1 regularization can improve recognition accuracy and receiver operator characteristic curves under noise corruption and occlusion.

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83. A Self-Organized Artificial Neural Network Architecture for Sensory Integration with Applications to Letter-Phoneme Integration

Tamas Jantvik, Lennart Gustafsson, Andrew P. Papli■ski

https://doi.org/10.1162/neco_a_00149 | pp. 2101-2139

The multimodal self-organizing network (MMSON), an artificial neural network architecture carrying out sensory integration, is presented here. The architecture is designed using neurophysiological findings and imaging studies that pertain to sensory integration and consists of interconnected lattices of artificial neurons. In this artificial neural architecture, the degree of recognition of stimuli, that is, the perceived reliability of stimuli in the various subnetworks, is included in the computation. The MMSON's behavior is compared to aspects of brain function that deal with sensory integration. According to human behavioral studies, integration of signals from sensory receptors of different modalities enhances perception of objects and events and also reduces time to detection. In neocortex, integration takes place in bimodal and multimodal association areas and result, not only in feedback-mediated enhanced unimodal perception and shortened reaction time, but also in robust bimodal or multimodal percepts. Simulation data from the presented artificial neural network architecture show that it replicates these important psychological and neuroscientific characteristics of sensory integration.

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84. Asymptotic Convergence Properties of the EM Algorithm for Mixture of Experts

Yan Yang, Jinwen Ma

https://doi.org/10.1162/neco_a_00154 | pp. 2140-2168

Mixture of experts (ME) is a modular neural network architecture for supervised classification. The double-loop expectation-maximization (EM) algorithm has been developed for learning the parameters of the ME architecture, and the iteratively reweighted least squares (IRLS) algorithm and the Newton-Raphson algorithm are two popular schemes for learning the parameters in the inner loop or gating network. In this letter, we investigate asymptotic convergence properties of the EM algorithm for ME using either the IRLS or Newton-Raphson approach. With the help of an overlap measure for the ME model, we obtain an upper bound of the asymptotic convergence rate of the EM algorithm in each case. Moreover, we find that for the Newton approach as a specific Newton-Raphson approach to learning the parameters in the inner loop, the upper bound of asymptotic convergence rate of the EM algorithm locally around the true solution Θ^* is [Formula: see text], where ■>0 is an arbitrarily small number, o(x) means that it is a higher-order infinitesimal as $x \to 0$, and $e(\Theta^*)$ is a measure of the average overlap of the ME model. That is, as the average overlap of the true ME model with large sample tends to zero, the EM algorithm with the Newton approach to learning the parameters in the inner loop tends to be asymptotically superlinear. Finally, we

substantiate our theoretical results by simulation experiments.

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85. Finding the Event Structure of Neuronal Spike Trains

J. Vincent Toups, Jean■Marc Fellous, Peter J. Thomas, Terrence J. Sejnowski, Paul Tiesinga
https://doi.org/10.1162/neco_a_00173 | pp. 2169-2208

Neurons in sensory systems convey information about physical stimuli in their spike trains. In vitro, single neurons respond precisely and reliably to the repeated injection of the same fluctuating current, producing regions of elevated firing rate, termed events. Analysis of these spike trains reveals that multiple distinct spike patterns can be identified as trial-to-trial correlations between spike times (Fellous, Tiesinga, Thomas, & Sejnowski, 2004). Finding events in data with realistic spiking statistics is challenging because events belonging to different spike patterns may overlap. We propose a method for finding spiking events that uses contextual information to disambiguate which pattern a trial belongs to. The procedure can be applied to spike trains of the same neuron across multiple trials to detect and separate responses obtained during different brain states. The procedure can also be applied to spike trains from multiple simultaneously recorded neurons in order to identify volleys of near-synchronous activity or to distinguish between excitatory and inhibitory neurons. The procedure was tested using artificial data as well as recordings in vitro in response to fluctuating current waveforms.

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86. The Dependence of Spike Field Coherence on Expected Intensity

Kyle Q. Lepage, Mark Kramer, Uri T. Eden
https://doi.org/10.1162/neco_a_00169 | pp. 2209-2241

The coherence between neural spike trains and local-field potential recordings, called spike-field coherence, is of key importance in many neuroscience studies. In this work, aside from questions of estimator performance, we demonstrate that theoretical spike-field coherence for a broad class of spiking models depends on the expected rate of spiking. This rate dependence confounds the phase locking of spike events to field-potential oscillations with overall neuron activity and is demonstrated analytically, for a large class of stochastic models, and in simulation. Finally, the relationship between the spike-field coherence and the intensity field coherence is detailed analytically. This latter quantity is independent of neuron firing rate and, under commonly found conditions, is proportional to the probability that a neuron spikes at a specific phase of field oscillation. Hence, intensity field coherence is a rate-independent measure and a candidate on which to base the appropriate statistical inference of spike field synchrony.

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87. Active Data Collection for Efficient Estimation and Comparison of Nonlinear Neural Models

Christopher DiMattina, Kechen Zhang
https://doi.org/10.1162/neco_a_00167 | pp. 2242-2288

The stimulus-response relationship of many sensory neurons is nonlinear, but fully quantifying this relationship by a complex nonlinear model may require too much data to be experimentally tractable. Here we present a theoretical study of a general two-stage computational method that may help to significantly

reduce the number of stimuli needed to obtain an accurate mathematical description of nonlinear neural responses. Our method of active data collection first adaptively generates stimuli that are optimal for estimating the parameters of competing nonlinear models and then uses these estimates to generate stimuli online that are optimal for discriminating these models. We applied our method to simple hierarchical circuit models, including nonlinear networks built on the spatiotemporal or spectral-temporal receptive fields, and confirmed that collecting data using our two-stage adaptive algorithm was far more effective for estimating and comparing competing nonlinear sensory processing models than standard nonadaptive methods using random stimuli.

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88. Invariant Object Recognition and Pose Estimation with Slow Feature Analysis

Mathias Franzius, Niko Wilbert, Laurenz Wiskott

https://doi.org/10.1162/neco_a_00171 | pp. 2289-2323

Primates are very good at recognizing objects independent of viewing angle or retinal position, and they outperform existing computer vision systems by far. But invariant object recognition is only one prerequisite for successful interaction with the environment. An animal also needs to assess an object's position and relative rotational angle. We propose here a model that is able to extract object identity, position, and rotation angles. We demonstrate the model behavior on complex three-dimensional objects under translation and rotation in depth on a homogeneous background. A similar model has previously been shown to extract hippocampal spatial codes from quasi-natural videos. The framework for mathematical analysis of this earlier application carries over to the scenario of invariant object recognition. Thus, the simulation results can be explained analytically even for the complex high-dimensional data we employed.

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89. A Differential Model of the Complex Cell

Miles Hansard, Radu Horaud

https://doi.org/10.1162/neco_a_00163 | pp. 2324-2357

The receptive fields of simple cells in the visual cortex can be understood as linear filters. These filters can be modeled by Gabor functions or gaussian derivatives. Gabor functions can also be combined in an energy model of the complex cell response. This letter proposes an alternative model of the complex cell, based on gaussian derivatives. It is most important to account for the insensitivity of the complex response to small shifts of the image. The new model uses a linear combination of the first few derivative filters, at a single position, to approximate the first derivative filter, at a series of adjacent positions. The maximum response, over all positions, gives a signal that is insensitive to small shifts of the image. This model, unlike previous approaches, is based on the scale space theory of visual processing. In particular, the complex cell is built from filters that respond to the 2D differential structure of the image. The computational aspects of the new model are studied in one and two dimensions, using the steerability of the gaussian derivatives. The response of the model to basic images, such as edges and gratings, is derived formally. The response to natural images is also evaluated, using statistical measures of shift insensitivity. The neural implementation and predictions of the model are discussed.

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90. Principles and Typical Computational Limitations of Sparse Speaker Separation Based on Deterministic Speech Features

A. Kern, Ruedi Stoop

https://doi.org/10.1162/neco_a_00165 | pp. 2358-2389

The separation of mixed auditory signals into their sources is an eminent neuroscience and engineering challenge. We reveal the principles underlying a deterministic, neural network-like solution to this problem. This approach is orthogonal to ICA/PCA that views the signal constituents as independent realizations of random processes. We demonstrate exemplarily that in the absence of salient frequency modulations, the decomposition of speech signals into local cosine packets allows for a sparse, noise-robust speaker separation. As the main result, we present analytical limitations inherent in the approach, where we propose strategies of how to deal with this situation. Our results offer new perspectives toward efficient noise cleaning and auditory signal separation and provide a new perspective of how the brain might achieve these tasks.

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91. Kernels for Longitudinal Data with Variable Sequence Length and Sampling Intervals

Zhengdong Lu, Todd K. Leen, Jeffrey Kaye

https://doi.org/10.1162/neco_a_00164 | pp. 2390-2420

We develop several kernel methods for classification of longitudinal data and apply them to detect cognitive decline in the elderly. We first develop mixed-effects models, a type of hierarchical empirical Bayes generative models, for the time series. After demonstrating their utility in likelihood ratio classifiers (and the improvement over standard regression models for such classifiers), we develop novel Fisher kernels based on mixture of mixed-effects models and use them in support vector machine classifiers. The hierarchical generative model allows us to handle variations in sequence length and sampling interval gracefully. We also give nonparametric kernels not based on generative models, but rather on the reproducing kernel Hilbert space. We apply the methods to detecting cognitive decline from longitudinal clinical data on motor and neuropsychological tests. The likelihood ratio classifiers based on the neuropsychological tests perform better than than classifiers based on the motor behavior. Discriminant classifiers performed better than likelihood ratio classifiers for the motor behavior tests.

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92. Algorithms for Nonnegative Matrix Factorization with the β-Divergence

Cédric Févotte, Jérôme Idier

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This letter describes algorithms for nonnegative matrix factorization (NMF) with the β-divergence (β-NMF). The β-divergence is a family of cost functions parameterized by a single shape parameter β that takes the Euclidean distance, the Kullback-Leibler divergence, and the Itakura-Saito divergence as special cases (β = 2, 1, 0 respectively). The proposed algorithms are based on a surrogate auxiliary function (a local majorization of the criterion function). We first describe a majorization-minimization algorithm that leads to multiplicative updates, which differ from standard heuristic multiplicative updates by a β-dependent power exponent. The monotonicity of the heuristic algorithm can, however, be proven for $\beta \in (0, 1)$ using the

proposed auxiliary function. Then we introduce the concept of the majorization-equalization (ME) algorithm, which produces updates that move along constant level sets of the auxiliary function and lead to larger steps than MM. Simulations on synthetic and real data illustrate the faster convergence of the ME approach. The letter also describes how the proposed algorithms can be adapted to two common variants of NMF: penalized NMF (when a penalty function of the factors is added to the criterion function) and convex NMF (when the dictionary is assumed to belong to a known subspace).

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93. A Systematic Method for Configuring VLSI Networks of Spiking Neurons

Emre Neftci, Elisabetta Chicca, Giacomo Indiveri, Rodney J. Douglas

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An increasing number of research groups are developing custom hybrid analog/digital very large scale integration (VLSI) chips and systems that implement hundreds to thousands of spiking neurons with biophysically realistic dynamics, with the intention of emulating brainlike real-world behavior in hardware and robotic systems rather than simply simulating their performance on general-purpose digital computers. Although the electronic engineering aspects of these emulation systems is proceeding well, progress toward the actual emulation of brainlike tasks is restricted by the lack of suitable high-level configuration methods of the kind that have already been developed over many decades for simulations on general-purpose computers. The key difficulty is that the dynamics of the CMOS electronic analogs are determined by transistor biases that do not map simply to the parameter types and values used in typical abstract mathematical models of neurons and their networks. Here we provide a general method for resolving this difficulty. We describe a parameter mapping technique that permits an automatic configuration of VLSI neural networks so that their electronic emulation conforms to a higher-level neuronal simulation. We show that the neurons configured by our method exhibit spike timing statistics and temporal dynamics that are the same as those observed in the software simulated neurons and, in particular, that the key parameters of recurrent VLSI neural networks (e.g., implementing soft winner-take-all) can be precisely tuned. The proposed method permits a seamless integration between software simulations with hardware emulations and intertranslatability between the parameters of abstract neuronal models and their emulation counterparts. Most important, our method offers a route toward a high-level task configuration language for neuromorphic VLSI systems.

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94. Characterization of Minimum Error Linear Coding with Sensory and Neural Noise

Eizaburo Doi, Michael S. Lewicki

https://doi.org/10.1162/neco_a_00181 | pp. 2498-2510

Robust coding has been proposed as a solution to the problem of minimizing decoding error in the presence of neural noise. Many real-world problems, however, have degradation in the input signal, not just in neural representations. This generalized problem is more relevant to biological sensory coding where internal noise arises from limited neural precision and external noise from distortion of sensory signal such as blurring and phototransduction noise. In this note, we show that the optimal linear encoder for this problem can be decomposed exactly into two serial processes that can be optimized separately. One is Wiener filtering, which optimally compensates for input degradation. The other is robust coding, which best uses the

available representational capacity for signal transmission with a noisy population of linear neurons. We also present spectral analysis of the decomposition that characterizes how the reconstruction error is minimized under different input signal spectra, types and amounts of degradation, degrees of neural precision, and neural population sizes.

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95. An Optimal Feedback Control Framework for Grasping Objects with Position Uncertainty

Vassilios N. Christopoulos, Paul Schrater

https://doi.org/10.1162/neco_a_00180 | pp. 2511-2536

As we move, the relative location between our hands and objects changes in uncertain ways due to noisy motor commands and imprecise and ambiguous sensory information. The impressive capabilities humans display for interacting and manipulating objects with position uncertainty suggest that our brain maintains representations of location uncertainty and builds compensation for uncertainty into its motor control strategies. Our previous work demonstrated that specific control strategies are used to compensate for location uncertainty. However, it is an open question whether compensation for position uncertainty in grasping is consistent with the stochastic optimal feedback control, mainly due to the difficulty of modeling natural tasks within this framework. In this study, we develop a stochastic optimal feedback control model to evaluate the optimality of human grasping strategies. We investigate the properties of the model through a series of simulation experiments and show that it explains key aspects of previously observed compensation strategies. It also provides a basis for individual differences in terms of differential control costs-the controller compensates only to the extent that performance benefits in terms of making stable grasps outweigh the additional control costs of compensation. These results suggest that stochastic optimal feedback control can be used to understand uncertainty compensation in complex natural tasks like grasping.

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96. A General Likelihood Framework for Characterizing the Time Course of Neural Activity

Michael J. Prerau, Uri T. Eden

https://doi.org/10.1162/neco_a_00185 | pp. 2537-2566

We develop a general likelihood-based framework for use in the estimation of neural firing rates, which is designed to choose the temporal smoothing parameters that maximize the likelihood of missing data. This general framework is algorithm-independent and thus can be applied to a multitude of established methods for firing rate or conditional intensity estimation. As a simple example of the use of the general framework, we apply it to the peristimulus time histogram and kernel smoother, the methods most widely used for firing rate estimation in the electrophysiological literature and practice. In doing so, we illustrate how the use of the framework can employ the general point process likelihood as a principled cost function and can provide substantial improvements in estimation accuracy for even the most basic of rate estimation algorithms. In particular, the resultant kernel smoother is simple to implement, efficient to compute, and can accurately determine the bandwidth of a given rate process from individual spike trains. We perform a simulation study to illustrate how the likelihood framework enables the kernel smoother to pick the bandwidth parameter that best predicts missing data, and we show applications to real experimental spike train data. Additionally, we discuss how the general likelihood framework may be used in conjunction with more sophisticated methods

for firing rate and conditional intensity estimation and suggest possible applications.

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97. Learning a Sparse Code for Temporal Sequences Using STDP and Sequence Compression

Sean Byrnes, Anthony N. Burkitt, David B. Grayden, Hamish Meffin

https://doi.org/10.1162/neco_a_00184 | pp. 2567-2598

A spiking neural network that learns temporal sequences is described. A sparse code in which individual neurons represent sequences and subsequences enables multiple sequences to be stored without interference. The network is founded on a model of sequence compression in the hippocampus that is robust to variation in sequence element duration and well suited to learn sequences through spike-timing dependent plasticity (STDP). Three additions to the sequence compression model underlie the sparse representation: synapses connecting the neurons of the network that are subject to STDP, a competitive plasticity rule so that neurons specialize to individual sequences, and neural depolarization after spiking so that neurons have a memory. The response to new sequence elements is determined by the neurons that have responded to the previous subsequence, according to the competitively learned synaptic connections. Numerical simulations show that the model can learn sets of intersecting sequences, presented with widely differing frequencies, with elements of varying duration.

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98. Automated Parameter Estimation of the Hodgkin-Huxley Model Using the Differential Evolution Algorithm: Application to Neuromimetic Analog Integrated Circuits

Laure Buhry, Filippo Grassia, Audrey Giremus, Éric Grivel, Sylvie Renaud, Sylvain Saïghi

https://doi.org/10.1162/neco_a_00170 | pp. 2599-2625

We propose a new estimation method for the characterization of the Hodgkin-Huxley formalism. This method is an alternative technique to the classical estimation methods associated with voltage clamp measurements. It uses voltage clamp type recordings, but is based on the differential evolution algorithm. The parameters of an ionic channel are estimated simultaneously, such that the usual approximations of classical methods are avoided and all the parameters of the model, including the time constant, can be correctly optimized. In a second step, this new estimation technique is applied to the automated tuning of neuromimetic analog integrated circuits designed by our research group. We present a tuning example of a fast spiking neuron, which reproduces the frequency-current characteristics of the reference data, as well as the membrane voltage behavior. The final goal of this tuning is to interconnect neuromimetic chips as neural networks, with specific cellular properties, for future theoretical studies in neuroscience.

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99. A Low-Order Model of Biological Neural Networks

James Lo

https://doi.org/10.1162/neco_a_00166 | pp. 2626-2682

A biologically plausible low-order model (LOM) of biological neural networks is proposed. LOM is a recurrent hierarchical network of models of dendritic nodes and trees; spiking and nonspiking neurons; unsupervised,

supervised covariance and accumulative learning mechanisms; feedback connections; and a scheme for maximal generalization. These component models are motivated and necessitated by making LOM learn and retrieve easily without differentiation, optimization, or iteration, and cluster, detect, and recognize multiple and hierarchical corrupted, distorted, and occluded temporal and spatial patterns. Four models of dendritic nodes are given that are all described as a hyperbolic polynomial that acts like an exclusive-OR logic gate when the model dendritic nodes input two binary digits. A model dendritic encoder that is a network of model dendritic nodes encodes its inputs such that the resultant codes have an orthogonality property. Such codes are stored in synapses by unsupervised covariance learning, supervised covariance learning, or unsupervised accumulative learning, depending on the type of postsynaptic neuron. A masking matrix for a dendritic tree, whose upper part comprises model dendritic encoders, enables maximal generalization on corrupted, distorted, and occluded data. It is a mathematical organization and idealization of dendritic trees with overlapped and nested input vectors. A model nonspiking neuron transmits inhibitory graded signals to modulate its neighboring model spiking neurons. Model spiking neurons evaluate the subjective probability distribution (SPD) of the labels of the inputs to model dendritic encoders and generate spike trains with such SPDs as firing rates. Feedback connections from the same or higher layers with different numbers of unit-delay devices reflect different signal traveling times, enabling LOM to fully utilize temporally and spatially associated information. Biological plausibility of the component models is discussed. Numerical examples are given to demonstrate how LOM operates in retrieving, generalizing, and unsupervised and supervised learning.

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100. Predicting Panel Data Binary Choice with the Gibbs Posterior

Lili Yao, Wenxin Jiang, Martin A. Tanner

https://doi.org/10.1162/neco_a_00172 | pp. 2683-2712

This letter considers Bayesian binary classification where data are assumed to consist of multiple time series (panel data) with binary class labels (binary choice). The observed data can be represented as {y it , x it } T, t=1 i = 1, … , n. Here y it ∈ {0, 1} represents binary choices, and x it represents the exogenous variables. We consider prediction of y it by its own lags, as well as by the exogenous components. The prediction will be based on a Bayesian treatment using a Gibbs posterior that is constructed directly from the empirical error of classification. Therefore, this approach is less sensitive to the misspecification of the probability model compared to the usual likelihood-based posterior, which is confirmed by Monte Carlo simulations. We also study the effects of various choices of n and T both numerically (by simulations) and theoretically (by considering two alternative asymptotic situations: large n and large T). We find that increasing T helps to reduce the prediction error more effectively compared to increasing n. We also illustrate the method in a real data application on the brand choice of yogurt purchases.

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101. Reproducing Kernel Banach Spaces with the ■¹Norm II: Error Analysis for Regularized Least Square Regression

Guohui Song, Haizhang Zhang

https://doi.org/10.1162/neco_a_00178 | pp. 2713-2729

A typical approach in estimating the learning rate of a regularized learning scheme is to bound the approximation error by the sum of the sampling error, the hypothesis error, and the regularization error.

Using a reproducing kernel space that satisfies the linear representer theorem brings the advantage of discarding the hypothesis error from the sum automatically. Following this direction, we illustrate how reproducing kernel Banach spaces with the ■ 1 norm can be applied to improve the learning rate estimate of ■ 1 -regularization in machine learning.

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102. Computing Confidence Intervals for Point Process Models

Sridevi V. Sarma, David P. Nguyen, Gabriela Czanner, Sylvia Wirth, Matthew Wilson, Wendy Suzuki, Emery N. Brown

https://doi.org/10.1162/neco_a_00198 | pp. 2731-2745

Characterizing neural spiking activity as a function of intrinsic and extrinsic factors is important in neuroscience. Point process models are valuable for capturing such information; however, the process of fully applying these models is not always obvious. A complete model application has four broad steps: specification of the model, estimation of model parameters given observed data, verification of the model using goodness of fit, and characterization of the model using confidence bounds. Of these steps, only the first three have been applied widely in the literature, suggesting the need to dedicate a discussion to how the time-rescaling theorem, in combination with parametric bootstrap sampling, can be generally used to compute confidence bounds of point process models. In our first example, we use a generalized linear model of spiking propensity to demonstrate that confidence bounds derived from bootstrap simulations are consistent with those computed from closed-form analytic solutions. In our second example, we consider an adaptive point process model of hippocampal place field plasticity for which no analytical confidence bounds can be derived. We demonstrate how to simulate bootstrap samples from adaptive point process models, how to use these samples to generate confidence bounds, and how to statistically test the hypothesis that neural representations at two time points are significantly different. These examples have been designed as useful guides for performing scientific inference based on point process models.

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103. The Combinatorics of Neurite Self-Avoidance

Elizabeth M. Forbes, Jonathan J. Hunt, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_00186 | pp. 2746-2769

During neural development in Drosophila, the ability of neurite branches to recognize whether they are from the same or different neurons depends crucially on the molecule Dscam1. In particular, this recognition depends on the stochastic acquisition of a unique combination of Dscam1 isoforms out of a large set of possible isoforms. To properly interpret these findings, it is crucial to understand the combinatorics involved, which has previously been attempted only using stochastic simulations for some specific parameter combinations. Here we present closed-form solutions for the general case. These reveal the relationships among the key variables and how these constrain possible biological scenarios.

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104. Self-Organization of Topographic Bilinear Networks for Invariant Recognition

Urs Bergmann, Christoph von der Malsburg

https://doi.org/10.1162/neco_a_00195 | pp. 2770-2797

We present a model for the emergence of ordered fiber projections that may serve as a basis for invariant recognition. After invariance transformations are self-organized, so-called control units competitively activate fiber projections for different transformation parameters. The model builds on a well-known ontogenetic mechanism, activity-based development of retinotopy, and it employs activity blobs of varying position and size to install different transformations. We provide a detailed analysis for the case of 1D input and output fields for schematic input patterns that shows how the model is able to develop specific mappings. We discuss results that show that the proposed learning scheme is stable for complex, biologically more realistic input patterns. Finally, we show that the model generalizes to 2D neuronal fields driven by simulated retinal waves.

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105. Reward-Weighted Regression with Sample Reuse for Direct Policy Search in Reinforcement Learning

Hirotaka Hachiya, Jan Peters, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00199 | pp. 2798-2832

Direct policy search is a promising reinforcement learning framework, in particular for controlling continuous, high-dimensional systems. Policy search often requires a large number of samples for obtaining a stable policy update estimator, and this is prohibitive when the sampling cost is expensive. In this letter, we extend an expectation-maximization-based policy search method so that previously collected samples can be efficiently reused. The usefulness of the proposed method, reward-weighted regression with sample reuse (R3), is demonstrated through robot learning experiments. (This letter is an extended version of our earlier conference paper: Hachiya, Peters, & Sugiyama, 2009 .).

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106. Estimating Parameters of Generalized Integrate-and-Fire Neurons from the Maximum Likelihood of Spike Trains

Yi Dong, ■tefan Mihala■, Alexander F. Russell, Ralph Etienne■Cummings, Ernst Niebur

https://doi.org/10.1162/neco_a_00196 | pp. 2833-2867

When a neuronal spike train is observed, what can we deduce from it about the properties of the neuron that generated it? A natural way to answer this question is to make an assumption about the type of neuron, select an appropriate model for this type, and then choose the model parameters as those that are most likely to generate the observed spike train. This is the maximum likelihood method. If the neuron obeys simple integrate-and-fire dynamics, Paninski, Pillow, and Simoncelli (2004) showed that its negative log-likelihood function is convex and that, at least in principle, its unique global minimum can thus be found by gradient descent techniques. Many biological neurons are, however, known to generate a richer repertoire of spiking behaviors than can be explained in a simple integrate-and-fire model. For instance, such a model retains only an implicit (through spike-induced currents), not an explicit, memory of its input; an example of a physiological situation that cannot be explained is the absence of firing if the input current is increased very slowly. Therefore, we use an expanded model (Mihalas & Niebur, 2009), which is capable of generating a large number of complex firing patterns while still being linear. Linearity is important because it maintains the distribution of the random variables and still allows maximum likelihood methods to be used.

In this study, we show that although convexity of the negative log-likelihood function is not guaranteed for this model, the minimum of this function yields a good estimate for the model parameters, in particular if the noise level is treated as a free parameter. Furthermore, we show that a nonlinear function minimization method (r-algorithm with space dilation) usually reaches the global minimum.

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107. A Model of Motion Transparency Processing with Local Center-Surround Interactions and Feedback

Florian Raudies, Ennio Mingolla, Heiko Neumann

https://doi.org/10.1162/neco_a_00193 | pp. 2868-2914

Motion transparency occurs when multiple coherent motions are perceived in one spatial location. Imagine, for instance, looking out of the window of a bus on a bright day, where the world outside the window is passing by and movements of passengers inside the bus are reflected in the window. The overlay of both motions at the window leads to motion transparency, which is challenging to process. Noisy and ambiguous motion signals can be reduced using a competition mechanism for all encoded motions in one spatial location. Such a competition, however, leads to the suppression of multiple peak responses that encode different motions, as only the strongest response tends to survive. As a solution, we suggest a local center-surround competition for population-encoded motion directions and speeds. Similar motions are supported, and dissimilar ones are separated, by representing them as multiple activations, which occurs in the case of motion transparency. Psychophysical findings, such as motion attraction and repulsion for motion transparency displays, can be explained by this local competition. Besides this local competition mechanism, we show that feedback signals improve the processing of motion transparency. A discrimination task for transparent versus opaque motion is simulated, where motion transparency is generated by superimposing large field motion patterns of either varying size or varying coherence of motion. The model's perceptual thresholds with and without feedback are calculated. We demonstrate that initially weak peak responses can be enhanced and stabilized through modulatory feedback signals from higher stages of processing.

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108. Synchronization and Redundancy: Implications for Robustness of Neural Learning and Decision Making

Jake Bouvrie, Jean-Jacques Slotine

https://doi.org/10.1162/neco_a_00183 | pp. 2915-2941

Learning and decision making in the brain are key processes critical to survival, and yet are processes implemented by nonideal biological building blocks that can impose significant error. We explore quantitatively how the brain might cope with this inherent source of error by taking advantage of two ubiquitous mechanisms, redundancy and synchronization. In particular we consider a neural process whose goal is to learn a decision function by implementing a nonlinear gradient dynamics. The dynamics, however, are assumed to be corrupted by perturbations modeling the error, which might be incurred due to limitations of the biology, intrinsic neuronal noise, and imperfect measurements. We show that error, and the associated uncertainty surrounding a learned solution, can be controlled in large part by trading off synchronization strength among multiple redundant neural systems against the noise amplitude. The impact of the coupling between such redundant systems is quantified by the spectrum of the network Laplacian,

and we discuss the role of network topology in synchronization and in reducing the effect of noise. We discuss range of situations in which the mechanisms we model arise in brain science and draw attention to experimental evidence suggesting that cortical circuits capable of implementing the computations of interest here can be found on several scales. Finally, simulations comparing theoretical bounds to the relevant empirical quantities show that the theoretical estimates we derive can be tight.

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109. Dependency Reduction with Divisive Normalization: Justification and Effectiveness

Siwei Lyu

https://doi.org/10.1162/neco_a_00197 | pp. 2942-2973

Efficient coding transforms that reduce or remove statistical dependencies in natural sensory signals are important for both biology and engineering. In recent years, divisive normalization (DN) has been advocated as a simple and effective nonlinear efficient coding transform. In this work, we first elaborate on the theoretical justification for DN as an efficient coding transform. Specifically, we use the multivariate t model to represent several important statistical properties of natural sensory signals and show that DN approximates the optimal transforms that eliminate statistical dependencies in the multivariate t model. Second, we show that several forms of DN used in the literature are equivalent in their effects as efficient coding transforms. Third, we provide a quantitative evaluation of the overall dependency reduction performance of DN for both the multivariate t models and natural sensory signals. Finally, we find that statistical dependencies in the multivariate t model and natural sensory signals are increased by the DN transform with low-input dimensions. This implies that for DN to be an effective efficient coding transform, it has to pool over a sufficiently large number of inputs.

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110. A Spline Framework for Estimating the EEG Surface Laplacian Using the Euclidean Metric

C. G. Carvalhaes, Patrick Suppes

https://doi.org/10.1162/neco_a_00192 | pp. 2974-3000

This letter develops a framework for EEG analysis and similar applications based on polyharmonic splines. This development overcomes a basic problem with the method of splines in the Euclidean setting: that it does not work on low-degree algebraic surfaces such as spherical and ellipsoidal scalp models. The method's capability is illustrated through simulations on the three-sphere model and using empirical data.

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111. Reappraisal of the Somatosensory Homunculus and Its Discontinuities

Pasha Parpia

https://doi.org/10.1162/neco_a_00179 | pp. 3001-3015

Neuroscience folklore has it that somatotopy in human primary somatosensory cortex (SI) has two significant discontinuities: the hands and face map onto adjacent regions in SI, as do the feet and genitalia. It has been proposed that these conjunctions in SI result from coincident sources of stimulation in the fetal position, where the hands frequently touch the face, and the feet the genitalia. Computer modeling using a

Hebbian variant of the self-organizing Kohonen net is consistent with this proposal. However, recent work reveals that the genital representation in SI for cutaneous sensations (as opposed to tumescence) is continuous with that of the lower trunk and thigh. This result, in conjunction with reports of separate face innervation and its earlier onset of sensory function, compared to that of the rest of the body, allows a reappraisal of homuncular organization. It is proposed that the somatosensory homunculus comprises two distinct somatotopic regions: the face representation and that of the rest of the body. Principles of self-organization do not account satisfactorily for the overall homuncular map. These results may serve to alert computational modelers that intrinsic developmental factors can override simple rules of plasticity.

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112. Improved Similarity Measures for Small Sets of Spike Trains

Richard Naud, Felipe Gerhard, Skander Mensi, Wulfram Gerstner

https://doi.org/10.1162/neco_a_00208 | pp. 3016-3069

Multiple measures have been developed to quantify the similarity between two spike trains. These measures have been used for the quantification of the mismatch between neuron models and experiments as well as for the classification of neuronal responses in neuroprosthetic devices and electrophysiological experiments. Frequently only a few spike trains are available in each class. We derive analytical expressions for the small-sample bias present when comparing estimators of the time-dependent firing intensity. We then exploit analogies between the comparison of firing intensities and previously used spike train metrics and show that improved spike train measures can be successfully used for fitting neuron models to experimental data, for comparisons of spike trains, and classification of spike train data. In classification tasks, the improved similarity measures can increase the recovered information. We demonstrate that when similarity measures are used for fitting mathematical models, all previous methods systematically underestimate the noise. Finally, we show a striking implication of this deterministic bias by reevaluating the results of the single-neuron prediction challenge.

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113. Estimation of Time-Dependent Input from Neuronal Membrane Potential

Ryota Kobayashi, Shigeru Shinomoto, Petr Lánský

https://doi.org/10.1162/neco_a_00205 | pp. 3070-3093

The set of firing rates of the presynaptic excitatory and inhibitory neurons constitutes the input signal to the postsynaptic neuron. Estimation of the time-varying input rates from intracellularly recorded membrane potential is investigated here. For that purpose, the membrane potential dynamics must be specified. We consider the Ornstein-Uhlenbeck stochastic process, one of the most common single-neuron models, with time-dependent mean and variance. Assuming the slow variation of these two moments, it is possible to formulate the estimation problem by using a state-space model. We develop an algorithm that estimates the paths of the mean and variance of the input current by using the empirical Bayes approach. Then the input firing rates are directly available from the moments. The proposed method is applied to three simulated data examples: constant signal, sinusoidally modulated signal, and constant signal with a jump. For the constant signal, the estimation performance of the method is comparable to that of the traditionally applied maximum likelihood method. Further, the proposed method accurately estimates both continuous and discontinuous time-variable signals. In the case of the signal with a jump, which does not satisfy the assumption of slow variability, the robustness of the method is verified. It can be concluded that the method provides reliable

estimates of the total input firing rates, which are not experimentally measurable.

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114. Identification and Continuity of the Distributions of Burst-Length and Interspike Intervals in the Stochastic Morris-Lecar Neuron

Peter F. Rowat, Priscilla E. Greenwood

https://doi.org/10.1162/neco_a_00209 | pp. 3094-3124

Using the Morris-Lecar model neuron with a type II parameter set and K(+)-channel noise, we investigate the interspike interval distribution as increasing levels of applied current drive the model through a subcritical Hopf bifurcation. Our goal is to provide a quantitative description of the distributions associated with spiking as a function of applied current. The model generates bursty spiking behavior with sequences of random numbers of spikes (bursts) separated by interburst intervals of random length. This kind of spiking behavior is found in many places in the nervous system, most notably, perhaps, in stuttering inhibitory interneurons in cortex. Here we show several practical and inviting aspects of this model, combining analysis of the stochastic dynamics of the model with estimation based on simulations. We show that the parameter of the exponential tail of the interspike interval distribution is in fact continuous over the entire range of plausible applied current, regardless of the bifurcations in the phase portrait of the model. Further, we show that the spike sequence length, apparently studied for the first time here, has a geometric distribution whose associated parameter is continuous as a function of applied current over the entire input range. Hence, this model is applicable over a much wider range of applied current than has been thought.

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115. Optimizing Time Histograms for Non-Poissonian Spike Trains

Takahiro Omi, Shigeru Shinomoto

https://doi.org/10.1162/neco_a_00213 | pp. 3125-3144

The time histogram is a fundamental tool for representing the inhomogeneous density of event occurrences such as neuronal firings. The shape of a histogram critically depends on the size of the bins that partition the time axis. In most neurophysiological studies, however, researchers have arbitrarily selected the bin size when analyzing fluctuations in neuronal activity. A rigorous method for selecting the appropriate bin size was recently derived so that the mean integrated squared error between the time histogram and the unknown underlying rate is minimized (Shimazaki & Shinomoto, 2007). This derivation assumes that spikes are independently drawn from a given rate. However, in practice, biological neurons express non-Poissonian features in their firing patterns, such that the spike occurrence depends on the preceding spikes, which inevitably deteriorate the optimization. In this letter, we revise the method for selecting the bin size by considering the possible non-Poissonian features. Improvement in the goodness of fit of the time histogram is assessed and confirmed by numerically simulated non-Poissonian spike trains derived from the given fluctuating rate. For some experimental data, the revised algorithm transforms the shape of the time histogram from the Poissonian optimization method.

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116. Learning Rule of Homeostatic Synaptic Scaling: Presynaptic Dependent or Not

Jian K. Liu
https://doi.org/10.1162/neco_a_00210 | pp. 3145-3161

It has been established that homeostatic synaptic scaling plasticity can maintain neural network activity in a stable regime. However, the underlying learning rule for this mechanism is still unclear. Whether it is dependent on the presynaptic site remains a topic of debate. Here we focus on two forms of learning rules: traditional synaptic scaling (SS) without presynaptic effect and presynaptic-dependent synaptic scaling (PSD). Analysis of the synaptic matrices reveals that transition matrices between consecutive synaptic matrices are distinct: they are diagonal and linear to neural activity under SS, but become nondiagonal and nonlinear under PSD. These differences produce different dynamics in recurrent neural networks. Numerical simulations show that network dynamics are stable under PSD but not SS, which suggests that PSD is a better form to describe homeostatic synaptic scaling plasticity. Matrix analysis used in the study may provide a novel way to examine the stability of learning dynamics.

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117. Adaptive Decoding for Brain-Machine Interfaces Through Bayesian Parameter Updates

Zheng Li, Joseph E. O'Doherty, Mikhail Lebedev, Miguel A. L. Nicolelis
https://doi.org/10.1162/neco_a_00207 | pp. 3162-3204

Brain-machine interfaces (BMIs) transform the activity of neurons recorded in motor areas of the brain into movements of external actuators. Representation of movements by neuronal populations varies over time, during both voluntary limb movements and movements controlled through BMIs, due to motor learning, neuronal plasticity, and instability in recordings. To ensure accurate BMI performance over long time spans, BMI decoders must adapt to these changes. We propose the Bayesian regression self-training method for updating the parameters of an unscented Kalman filter decoder. This novel paradigm uses the decoder's output to periodically update its neuronal tuning model in a Bayesian linear regression. We use two previously known statistical formulations of Bayesian linear regression: a joint formulation, which allows fast and exact inference, and a factorized formulation, which allows the addition and temporary omission of neurons from updates but requires approximate variational inference. To evaluate these methods, we performed offline reconstructions and closed-loop experiments with rhesus monkeys implanted cortically with microwire electrodes. Offline reconstructions used data recorded in areas M1, S1, PMd, SMA, and PP of three monkeys while they controlled a cursor using a handheld joystick. The Bayesian regression self-training updates significantly improved the accuracy of offline reconstructions compared to the same decoder without updates. We performed 11 sessions of real-time, closed-loop experiments with a monkey implanted in areas M1 and S1. These sessions spanned 29 days. The monkey controlled the cursor using the decoder with and without updates. The updates maintained control accuracy and did not require information about monkey hand movements, assumptions about desired movements, or knowledge of the intended movement goals as training signals. These results indicate that Bayesian regression self-training can maintain BMI control accuracy over long periods, making clinical neuroprosthetics more viable.

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118. Subthreshold Membrane Depolarization as Memory Trace for Perceptual Learning

Osamu Hoshino

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Experience-dependent synaptic plasticity characterizes the adaptable brain and is believed to be the cellular substrate for perceptual learning. A chemical agent such as gamma-aminobutyric acid (GABA) is known to affect synaptic alteration, perhaps gating perceptual learning. We examined whether and how ambient (extrasynaptic) GABA affects experience-dependent synaptic alteration. A cortical neural network model was simulated. Transporters on GABAergic interneurons regulate ambient GABA levels around their axonal target neurons by removing GABA from (forward transport) or releasing it into (reverse transport) the extracellular space. The ambient GABA provides neurons with tonic inhibitory currents by activating extrasynaptic GABA(a) receptors. During repeated exposures to the same stimulus, we modified the synaptic connection strength between principal cells in a spike-timing-dependent manner. This modulated the activity of GABAergic interneurons, and reduced or augmented ambient GABA concentration. Reduction in ambient GABA concentration led to slight depolarization (less than several millivolts) in ongoing-spontaneous membrane potential. This was a subthreshold neuronal behavior because ongoing-spontaneous spiking activity remained almost unchanged. The ongoing-spontaneous subthreshold depolarization improved a suprathreshold neuronal response. If the stimulus was long absent for perceptual learning, augmentation of ambient GABA concentration took place and the ongoing-spontaneous subthreshold depolarization was depressed. We suggest that a perceptual memory trace could be left in neuronal circuitry as an ongoing-spontaneous subthreshold membrane depolarization, which would allow that memory to be accessed easily afterward, whereas a trace of a memory that has not recently been retrieved fades away when the ongoing-spontaneous subthreshold membrane depolarization built by previous perceptual learning is depressed. This would lead that memory to be accessed with some difficulty. In the brain, ambient GABA, whose level could be regulated by transporter may have an important role in leaving memory trace for perceptual learning.

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119. Neural Mass Activity, Bifurcations, and Epilepsy

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In this letter, we propose a general framework for studying neural mass models defined by ordinary differential equations. By studying the bifurcations of the solutions to these equations and their sensitivity to noise, we establish an important relation, similar to a dictionary, between their behaviors and normal and pathological, especially epileptic, cortical patterns of activity. We then apply this framework to the analysis of two models that feature most phenomena of interest, the Jansen and Rit model, and the slightly more complex model recently proposed by Wendling and Chauvel. This model-based approach allows us to test various neurophysiological hypotheses on the origin of pathological cortical behaviors and investigate the effect of medication. We also study the effects of the stochastic nature of the inputs, which gives us clues about the origins of such important phenomena as interictal spikes, interictal bursts, and fast onset activity that are of particular relevance in epilepsy.

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120. On the Relation of Slow Feature Analysis and Laplacian Eigenmaps

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The past decade has seen a rise of interest in Laplacian eigenmaps (LEMs) for nonlinear dimensionality reduction. LEMs have been used in spectral clustering, in semisupervised learning, and for providing efficient state representations for reinforcement learning. Here, we show that LEMs are closely related to slow feature analysis (SFA), a biologically inspired, unsupervised learning algorithm originally designed for learning invariant visual representations. We show that SFA can be interpreted as a function approximation of LEMs, where the topological neighborhoods required for LEMs are implicitly defined by the temporal structure of the data. Based on this relation, we propose a generalization of SFA to arbitrary neighborhood relations and demonstrate its applicability for spectral clustering. Finally, we review previous work with the goal of providing a unifying view on SFA and LEMs.